My Learning Draft

Labix

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Contents

1 Combinatorial Commutative Algebra

Pre-req: Commutative Algebra 2, Simplicial Complexes

1.1 Stanley-Reisner Rings

Theorem 1.1.1
Let R be a commutative ring. Let $n \in \mathbb{N}$. There is a bijection

$$\begin{matrix}\text{Abstract Simplicial}\\ \text{Complexes on } \{1,\dots,n\}\end{matrix} \overset{1:1}{\leftrightarrow} \begin{matrix}\text{Squarefree monomial}\\ \text{ideals of } R[x_1,\dots,x_n]\end{matrix}$$

given by sending a simplicial set S to the ideal $\left\langle \prod_{j\in J} x_j \mid J \in \mathcal{P}(\{1,\dots,n\}) \setminus S \right\rangle$.

Definition 1.1.2 (Stanley-Reisner Rings)
Let R be a commutative ring. Let $n \in \mathbb{N}$. Let S be an abstract simplicial complex on $\{1,\dots,n\}$. Define the Stanley-Reisner ring of S to be

$$R[S] = \frac{k[x_1,\dots,x_n]}{\left\langle \prod_{j\in J} x_j \mid J \in \mathcal{P}(\{1,\dots,n\}) \setminus S \right\rangle}$$

The ideal $I_S = \left\langle \prod_{j\in J} x_j \mid J \in \mathcal{P}(\{1,\dots,n\}) \setminus S \right\rangle$ is called the Stanley-Reisner ideal of S.

Example 1.1.3
Let R be a commutative ring. The Stanley-Reisner ring of the standard n-simplex Δ^n is given by

$$R[\Delta^n] = k[x_1,\dots,x_n]$$

Example 1.1.4
Let R be a commutative ring. The Stanley-Reisner ring of the abstract simplicial complex $S = \mathcal{P}(\{1,2,3\}) \cup \mathcal{P}(\{1,3,4\}) \cup \mathcal{P}(\{2,4\})$ over $\{1,2,3,4\}$ is given by

$$R[S] = \frac{k[x_1,x_2,x_3,x_4]}{(x_1x_2x_4, x_2x_3x_4)}$$

Proposition 1.1.5
Let R be a commutative ring. Let $n \in \mathbb{N}$. Let S be an abstract simplicial complex on $\{1,\dots,n\}$. Then we have

$$I_S = \bigcap_{J\in S} \langle x_i \mid i \notin J \rangle$$

Maps of simplicial sets -> Maps of k-algebra.

Proposition 1.1.6
Let R be a commutative ring. Let $\mathcal{K}_1, \mathcal{K}_2$ be simplicial complexes. Then we have

$$R[\mathcal{K}_1 * \mathcal{K}_2] \cong R[\mathcal{K}_1] \otimes_R R[\mathcal{K}_2]$$

Proposition 1.1.7
Let R be a commutative ring. Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. If $\mathcal{K}$ is a flag complex, then $R[\mathcal{K}]$ is a Koszul algebra.

1.2 The Hilbert Series of Stanley-Reisner Rings

Theorem 1.2.1 (Stanley)
Let k be a field. Let $n \in \mathbb{N}$. Let S be an abstract simplicial complex on $\{1, \dots, n\}$. Let $(f_0, \dots, f_n)$ be the face vector of S. Let $(h_0, \dots, h_n)$ be the h-vector of S. Then the Hilbert series of $k[S]$ is given by

$$HS_{k[S]}(t) = \sum_{i=0}^{n+1} f_{k-1} \left(\frac{t^2}{1-t^2} \right)^k = \frac{1}{(1-t^2)^{n+1}} \sum_{i=0}^{n+1} h_i t^{2i}$$

Example 1.2.2 | Let Δ^n be the standard n-simplex. Then the following are true.

- The Hilbert series for the Stanley-Reisner ring of Δ^n is given by

$$HS_{k[\Delta^n]}(t) = \frac{1}{(1-t^2)^{n+1}}$$

- The Hilbert series for the Stanley-Reisner ring of $\partial\Delta^n$ is given by

$$HS_{k[\partial\Delta^n]}(t) = \frac{1}{(1-t^2)^{n+1}} \sum_{i=0}^{n+1} t^{2i}$$

2 Polyhedral Products

2.1 The Homotopy Theory of Polyhedral Products

Definition 2.1.1 (Polyhedral Products)
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Let $(\mathbf{X},\mathbf{A}) = \{(X_1,A_1),\dots,(X_n,A_n)\}$ be pairs of spaces.

- Let $\sigma\in\mathcal{K}$. Define
$$(\mathbf{X},\mathbf{A})^\sigma = \left\{(x_1,\dots,x_n)\in\prod_{i=1}^n X_i \;\middle|\; x_i\in A_i \text{ for } i\notin I\right\}$$
- Define the polyhedral product of $(\mathbf{X},\mathbf{A})$ corresponding to $\mathcal{K}$ to be
$$(\mathbf{X},\mathbf{A})^\mathcal{K} = \bigcup_{I\in\mathcal{K}}(\mathbf{X},\mathbf{A})^\sigma$$

Proposition 2.1.2
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Let $(\mathbf{X},\mathbf{A}) = \{(X_1,A_1),\dots,(X_n,A_n)\}$ be pairs of spaces. Then we have
$$(\mathbf{X},\mathbf{A})^\mathcal{K} = \operatorname*{colim}_{\sigma\in\text{Cat}(\mathcal{K})}(\mathbf{X},\mathbf{A})^\sigma$$

Proposition 2.1.3
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Let $(\mathbf{X},\mathbf{A})$ and $(\mathbf{Y},\mathbf{B})$ be n pairs of spaces. Let $f_i:(X_i,A_i)\to(Y_i,B_i)$ be a map of pair for $1\le i\le n)$. Then $(f_1,\dots,f_n)$ induces a map
$$(\mathbf{X},\mathbf{A})^\mathcal{K}\to(\mathbf{Y},\mathbf{B})^\mathcal{K}$$

Proposition 2.1.4
Let $\mathcal{K},\mathcal{L}$ be a simplicial complex on $\{1,\dots,n\}$. Let $(\mathbf{X},\mathbf{A}) = \{(X_1,A_1),\dots,(X_n,A_n)\}$ be pairs of spaces. Let $f_i:(X_i,A_i)\to(Y_i,B_i)$ be a map of pair for $1\le i\le n)$. Then we have
$$(\mathbf{X},\mathbf{A})^{\mathcal{K}*\mathcal{L}} = (\mathbf{X},\mathbf{A})^\mathcal{K}\times(\mathbf{X},\mathbf{A})^\mathcal{L}$$

Example 2.1.5
Let $\mathcal{K}$ be a simplicial complex. Then we have
$$\text{cc}(\mathcal{K}) = (I,1)^\mathcal{K}$$

Example 2.1.6
Let $\mathcal{K} = \{\{1,\},\dots,\{n\}\}$. Let $\mathbf{X} = (X_1,\dots,X_n)$ be spaces. Then we have
$$(\mathbf{X}X,*)^\mathcal{K} = X_1\vee\cdots\vee X_n$$

Definition 2.1.7 (Polyhedral Smash Products)
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Let $(\mathbf{X},\mathbf{A})$ be pairs of spaces.

- Let $\sigma\in\mathcal{K}$. Define
$$(\mathbf{X},\mathbf{A})^{\wedge\sigma} = \left\{(x_1,\dots,x_n)\in\bigwedge_{i=1}^n X_i \;\middle|\; x_i\in A_i \text{ for } i\notin I\right\}$$
- Define the polyhedral smash product of $(\mathbf{X},\mathbf{A})$ corresponding to $\mathcal{K}$ to be
$$(\mathbf{X},\mathbf{A})^\mathcal{K} = \bigcup_{I\in\mathcal{K}}(\mathbf{X},\mathbf{A})^{\wedge\sigma}$$

Theorem 2.1.8 (Bahri, Bendersky, Cohen, Gitler)
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Let $(\mathbf{X}, \mathbf{A})$ be n-pairs of CW complexes. Suppose that the inclusion maps $A_j \to X_j$ are null homotopic for $1 \leq j \leq n$. Then there is a homotopy equivalence

$$(\mathbf{X}, \mathbf{A})^{\wedge \mathcal{K}} \simeq \bigvee_{\sigma \in \mathcal{K}} |\mathrm{lk}_{\mathcal{K}}(\sigma)| * (\mathbf{X}_J, \mathbf{A}_J)^{\wedge \sigma}$$

Theorem 2.1.9 (Bahri, Bendersky, Cohen, Gitler)
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Let $(\mathbf{X}, \mathbf{A})$ be n-pairs of CW complexes. Then there is a homotopy equivalence

$$\Sigma(\mathbf{X}, \mathbf{A})^{\mathcal{K}} \simeq \Sigma \left(\bigvee_{J \subseteq \{1, \dots, n\}} (\mathbf{X}_J, \mathbf{A}_J)^{\wedge \mathcal{K}_J} \right)$$

that is natural in $(\mathbf{X}, \mathbf{A})$, where $(\mathbf{X}_J, \mathbf{A}_J) = \{(X_j, A_j) \mid j \in J\}$.

Corollary 2.1.10
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Let $(\mathbf{X}, \mathbf{A})$ be n-pairs of CW complexes. Suppose that A_i are null homotopic for all i. Then there is a homotopy equivalence

$$\Sigma(\mathbf{X}, \mathbf{A})^{\mathcal{K}} \simeq \Sigma \left(\bigvee_{\sigma \in \mathcal{K}} \mathbf{X}^{\wedge \sigma} \right)$$

Corollary 2.1.11
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Let $(\mathbf{X}, \mathbf{A})$ be n-pairs of CW complexes. Suppose that X_i are null homotopic for all i. Then there is a homotopy equivalence

$$\Sigma(\mathbf{X}, \mathbf{A})^{\mathcal{K}} \simeq \Sigma \left(\bigvee_{J \notin \mathcal{K}} |\mathcal{K}_J| * \mathbf{A}^{\wedge J} \right)$$

Theorem 2.1.12 (Grbic–Theriault, Iriye–Kishimoto)
Let $\mathcal{K}$ be a shifted simplicial complex on $\{1, \dots, n\}$. Let $\mathbf{A}$ be n pointed CW complexes. Then there is a homotopy equivalence

$$(CA, A)^{\mathcal{K}} \simeq \bigvee_{J \notin \mathcal{K}} |\mathcal{K}_J| * \mathbf{A}^{\wedge J}$$

Example 2.1.13
Let $\mathbf{A}$ be n pointed CW complexes. Then there is a homotopy equivalence

$$(CA, A)^{\mathrm{sk}_i \Delta^{n-1}} \simeq \bigvee_{k=i+2}^{n} \left(\bigvee_{1 \leq j_1 < \cdots < j_k \leq n} (\Sigma^{i+1} A_{j_1} \wedge \cdots \wedge A_{j_k})^{\vee \binom{k-1}{i+1}} \right)$$

2.2 Fat Wedge Filtrations

2.3 Other Results

Proposition 2.3.1
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then we have

$$H^*((\mathbb{CP}^\infty, *)^{\mathcal{K}}) \cong \mathbb{Z}[\mathcal{K}]$$

Proposition 2.3.2
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then the inclusion map $(\mathbb{CP}^\infty, *)^{\mathcal{K}} \hookrightarrow (\mathbb{CP}^\infty)^n$ induces the quotient map

$$\mathbb{Z}[x_1, \dots, x_n] \to \mathbb{Z}[\mathcal{K}]$$

where $\deg(x_i) = 2$.

3 Moment Angle Complexes

3.1 Basic Definitions

Definition 3.1.1 (Moment Angle Complexes)
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Define the moment angle complex of $\mathcal{K}$ to be

$$\mathcal{Z}_{\mathcal{K}} = (D^2, S^1)^{\mathcal{K}}$$

Proposition 3.1.2
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then the following diagram is a pullback

$$\begin{tikzcd} \mathcal{Z}_{\mathcal{K}} \arrow[r] \arrow[d] & (D^2)^n \subseteq \mathbb{C}^n \arrow[d, "\mu"] \\ \mathrm{cc}(\mathcal{K}) \arrow[r, hook] & I^n \end{tikzcd}$$

where $D^2 \subseteq \mathbb{C}$ is the closed unit disc and $\mu : (D^2)^n \to I^n$ is the map given by $(z_1, \dots, z_n) \mapsto (|z_1|^2, \dots, |z_n|^2)$.

Corollary 3.1.3
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then $\mathcal{Z}_{\mathcal{K}}$ is a T^n-equivariant subspace of $(D^2)^n$.

Example 3.1.4
The following are true.
- $\mathcal{Z}_{\Delta^{n-1}} = (D^2)^n$.
- $\mathcal{Z}_{\partial\Delta^{n-1}} \cong S^{2n-1}$.

Proposition 3.1.5
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$ that is a triangulation of S^m. Then $\mathcal{Z}_{\mathcal{K}}$ is a compact topological manifold of dimension $m + n + 1$.

3.2 Homotopy Theory of Moment Angle Complexes

Proposition 3.2.1
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then there is a factorization

$$\begin{tikzcd} (\mathbb{CP}^\infty, *)^{\mathcal{K}} \arrow[rr, hook] \arrow[dr, "h"'] & & (\mathbb{CP}^\infty)^n \\ & ET^n \times_{T^n} \mathcal{Z}_{\mathcal{K}} \arrow[ur, "p"'] & \end{tikzcd}$$

where h is a homotopy equivalence and p is a fibration with fiber $\mathcal{Z}_{\mathcal{K}}$.

Corollary 3.2.2
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then we have

$$H^*_{T^n}(\mathcal{Z}_{\mathcal{K}}) \cong \mathbb{Z}[\mathcal{K}]$$

Proposition 3.2.3
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then the following are true.
- $\mathcal{Z}_{\mathcal{K}}$ is 2-connected.
- $\pi_k(\mathcal{Z}_{\mathcal{K}}) \cong \pi_k((\mathbb{CP}^\infty, *)^{\mathcal{K}})$ for $k \geq 3$.

3.3 Cohomology of Moment Angle Complexes

Consider the decomposition of D^2 into the following CW complex structure:

- One 0-cell $1 \in D^2$.
- One 1-cell S^1.
- One 2-cell D^2.

Let $n \in \mathbb{N}$. This gives the product complex D^{2n} a CW complex structure, whose cells are a product of 1-cells and 2-cells. We parameterize the cells in the following way. For $J, I \subseteq \{1, \dots, n\}$ with $J \cap I = \emptyset$, we denote $\chi(J, I)$ as the $(|J| + |I|)$-cell corresponding to the cell with the factor of a 1-cell in the jth position for $j \in J$, the factor of a 0-cell in the ith position for $i \in I$, and the factor of a 0-cell for $k \notin I \cup J$.

Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then $\mathcal{Z}_\mathcal{K}$ is a union of cells in D^{2n}, and so is a cellular subcomplex of D^{2n}. Namely, for each $I \in \mathcal{K}$, we have that $\chi(J, I) \subseteq \mathcal{Z}_\mathcal{K}$ for all $J \subseteq \{1, \dots, n\} \setminus I$.

Q: How does the cellular cochain split into a bicomplex

Definition 3.3.1 (Bigrading in the Cochain Complex)
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Define a bigrading in $C^*(\mathcal{Z}_\mathcal{K}; \mathbb{Z})$ as follows. For each basis element $\chi(J, I)$, define $\deg(\chi(J, I)) = (-|J|, 2(|I| + |J|))$.

Lemma 3.3.2
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then the following are true.
- The $(2j)$th components for each $j \in \mathbb{N}$ defines a cochain complex $C^{-i,2j}(\mathcal{Z}_\mathcal{K}; \mathbb{Z})$ in i.
- $(C^{-*,*}(\mathcal{Z}_\mathcal{K}; \mathbb{Z}), d_{CW}, 0)$ is a double cochain complex.
- The cellular cochain complex of $\mathcal{Z}_\mathcal{K}$ splits into a direct sum
$$C^*(\mathcal{Z}_\mathcal{K}; \mathbb{Z}) = \text{Tot}^\oplus(C^{-*,*}(\mathcal{Z}_\mathcal{K})) = \bigoplus_{j \in \mathbb{N}} C^{-*,2j}(\mathcal{Z}_\mathcal{K}; \mathbb{Z})$$
where
$$C^k(\mathcal{Z}_\mathcal{K}; \mathbb{Z}) = \bigoplus_{2j-i=k} C^{-i,2j}(\mathcal{Z}_\mathcal{K}; \mathbb{Z})$$

The consequence of the splitting is that cohomology also accquires a splitting
$$H^n(\mathcal{Z}_\mathcal{K}; \mathbb{Z}) = \bigoplus_{2j-i=n} H^{-i}(C^{-*,2j}(\mathcal{Z}_\mathcal{K}; \mathbb{Z}))$$

Proposition 3.3.3
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then the following are true.
- There are natural isomorphisms
$$H^{-p}(C^{-*,2q}(\mathcal{Z}_\mathcal{K}; \mathbb{Z})) \cong H_{-p}\left(\Lambda\left(\bigoplus_{i=1}^n \mathbb{Z}[\mathcal{K}]u_i\right)\right)_{2q} = H_{-p}\left(\Lambda\left(\bigoplus_{i=1}^n \mathbb{Z}[\mathcal{K}]u_i\right)_{2q}\right)$$
where $\Lambda\left(\bigoplus_{i=1}^n R[\mathcal{K}]u_i\right)$ is the differential graded algebra whose homological grading is given by $\deg(u_i) = -1$, internal grading is given by $\deg(v_i) = 2$ and differential is given by $d(u_i) = v_i$ (and $d(v_i) = 0$ since we think of the DGA as a DGA over the ring $\mathbb{Z}[\mathcal{K}]$).
- There above natural isomorphisms induces natural isomorphisms
$$H^k(\mathcal{Z}_\mathcal{K}; \mathbb{Z}) \cong \bigoplus_{2q-p=k} H_{-p}\left(\Lambda\left(\bigoplus_{i=1}^n \mathbb{Z}[\mathcal{K}]u_i\right)\right)_{2q}$$
- The above natural isomorphisms induce a natural isomorphism of rings
$$H^*(\mathcal{Z}_\mathcal{K}; \mathbb{Z}) \cong H_*\left(\Lambda\left(\bigoplus_{i=1}^n \mathbb{Z}[\mathcal{K}]u_i\right)\right)$$

Note that $H^n(\mathcal{Z}_\mathcal{K};\mathbb{Z})$ is not isomorphic to $H_{-n}\left(\Lambda\left(\bigoplus_{i=1}^n \mathbb{Z}[\mathcal{K}]u_i\right)\right)$ despite the suggestive notation. This is a mismatch of algebraic objects since the left is an abelian group while the right hand side is a $\mathbb{Z}[\mathcal{K}]$-module. In order to have an isomorphism of abelian groups we must further decompose each homology $\mathbb{Z}[\mathcal{K}]$-modules further into $\mathbb{Z}$-modules by considering $\mathbb{Z}[\mathcal{K}]$ as a $\mathbb{Z}$-module. But this results in a shift of degrees.

Let $K_\bullet(v_1,\dots,v_n;\mathbb{Z}[\mathcal{K}])$ be the Koszul complex over $\mathbb{Z}[v_1,\dots,v_n]$. Notice that the cochain complex defined by $K^n = K_n(v_1,\dots,v_n;\mathbb{Z}[\mathcal{K}])$ is precisely our differential graded algebra $\Lambda\left(\bigoplus_{i=1}^n \mathbb{Z}[\mathcal{K}]u_i\right)$. Then the cohomology of $\mathcal{Z}_\mathcal{K}$ acquires an additional description:

$$H^n(\mathcal{Z}_\mathcal{K};\mathbb{Z}) \cong \bigoplus_{2q-p=n} \left(H_p\left(K_\bullet(v_1,\dots,v_n;\mathbb{Z}[\mathcal{K}])\right)\right)_{2q}$$

Finally, since $K_\bullet(v_1,\dots,v_n)$ is a free resolution for $\mathbb{Z}$ as a $\mathbb{Z}[v_1,\dots,v_n]$-module, we know that

$$H_p(K_\bullet(v_1,\dots,v_n;R[\mathcal{K}])) = \text{Tor}_p^{\mathbb{Z}[v_1,\dots,v_n]}(\mathbb{Z}[\mathcal{K}],\mathbb{Z})$$

The Koszul algebra gives the Tor groups the structure of an algebra, and so the cohomology ring acquires the description

$$H^*(\mathcal{Z}_\mathcal{K};\mathbb{Z}) \cong \text{Tor}_*^{\mathbb{Z}[v_1,\dots,v_n]}(\mathbb{Z}[v_1,\dots,v_n],\mathbb{Z})$$

But again note that the nth cohomology group of $\mathcal{Z}_\mathcal{K}$ is not isomorphic to the nth Tor group of $\mathbb{Z}[\mathcal{K}]$ and $\mathbb{Z}$ by virtually the same reason as above.

Corollary 3.3.4

Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then there is a homotopy equivalence

$$\Sigma\mathcal{Z}_\mathcal{K} \simeq \bigvee_{J\notin\mathcal{K}} \Sigma^{2+|J|}|\mathcal{K}_J|$$

Corollary 3.3.5

Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then the above homotopy equivalence descends to a natural isomorphism

$$\bigoplus_{J\subseteq\{1,\dots,n\}} C^{k-|J|-1}(\mathcal{K}_J;\mathbb{Z}) \cong C^*(\mathcal{Z}_\mathcal{K};\mathbb{Z})$$

given by

$$\alpha_L \mapsto \varepsilon(L,J)\chi(J\setminus L, L)$$

where α_L is the cochain basis of Hochster's formula given by the simplex $L\subseteq J$ and

$$\varepsilon(L,J) = \prod_{j\in L}\varepsilon(j,J)$$

and $\varepsilon(j,J) = (-1)^{r-1}$ if j is in the rth position of J.

Proposition 3.3.6

Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then the following are true.

- There above natural isomorphisms induces natural isomorphisms

$$H^{-i}(C^{-*,2j}(\mathcal{Z}_\mathcal{K};\mathbb{Z})) \cong \bigoplus_{\substack{J\subseteq\{1,\dots,n\}\\ |J|=j}} \widetilde{H}^{j-i-1}(\mathcal{K}_J;\mathbb{Z})$$

  where by convention we set $\widetilde{H}^{-1}(\mathcal{K}_\emptyset) = \mathbb{Z}$.
- The above natural isomorphisms induces natural isomorphisms

$$H^k(\mathcal{Z}_\mathcal{K};\mathbb{Z}) \cong \bigoplus_{J\subseteq\{1,\dots,n\}} \widetilde{H}^{k-|J|-1}(\mathcal{K}_J;\mathbb{Z})$$

We are missing a description of the cup product in terms of Hochster's formula. It is given by the following definition.

Definition 3.3.7 (Ring Structure in Hochster's Formula)
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Define a ring structure in $\bigoplus_{J\subseteq\{1,\dots,n\}} \widetilde{H}^*(\mathcal{K}_{\mathcal{J}};\mathbb{Z})$ as follows. Let α_L be the cochain basis element in $C^p(\mathcal{K}_I;\mathbb{Z})$ corresponding to the simplex $L\subseteq I$. The ring structure is induced by the map

$$C^p(\mathcal{K}_I;\mathbb{Z})\otimes_{\mathbb{Z}} C^q(\mathcal{K}_{\mathcal{J}};\mathbb{Z})\to C^{p+q+1}(\mathcal{K}_{I\cup J})$$

defined by

$$\alpha_L\otimes\alpha_M\mapsto\begin{cases}\varepsilon(L,I)\varepsilon(M,J)\varepsilon(L\cup M,I\cup J)\alpha_{L\amalg M} & \text{if } I\cap J=\emptyset\\ 0 & \text{otherwise}\end{cases}$$

where $\varepsilon(L,I)=\prod_{j\in L}\varepsilon(j,J)$ and $\varepsilon(j,J)=(-1)^{r-1}$ if j is in the rth position in J.

Proposition 3.3.8
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then the following are true.

- The ring structure in Hochster's formula is equal to the relative Kunneth theorem for

$$H^*(|\mathcal{K}_I*\mathcal{K}_J|)=H^*(\Sigma(|\mathcal{K}_I|\wedge|\mathcal{K}_J|))=\widetilde{H}^{*-1}(|K_I|\times|K_J|,|K_I\vee K_J|)$$

  via the inclusion $\mathcal{K}_{I\amalg J}\hookrightarrow\mathcal{K}_I*\mathcal{K}_J$.
- The natural isomorphism in the above prp induces a natural isomorphism of rings

$$H^*(\mathcal{Z}_{\mathcal{K}};\mathbb{Z})\cong\bigoplus_{J\subseteq\{1,\dots,n\}}\widetilde{H}^*(\mathcal{K}_J;\mathbb{Z})$$

While the cohomology groups can be easily computed using Hochster's formula, the ring structure can be seen more easily using the DGA.

Example 3.3.9
Let $\mathcal{K}$ be the standard polygon with vertices given by $\{1,\dots,5\}$ and edges given by $\{\{1,2\},\dots,\{4,5\},\{1,5\}\}$. We compute that

- $H^0(\mathcal{Z}_{\mathcal{K}};\mathbb{Z})=\bigoplus_{J\subseteq\{1,\dots,5\}}\widetilde{H}^{0-|J|-1}(\mathcal{K}_{\mathcal{J}};\mathbb{Z})=\widetilde{H}^{-1-|\emptyset|}(\mathcal{K}_\emptyset;\mathbb{Z})=\mathbb{Z}$.
- $H^1(\mathcal{Z}_{\mathcal{K}};\mathbb{Z})=H^2(\mathcal{Z}_{\mathcal{K}};\mathbb{Z})=H^5(\mathcal{Z}_{\mathcal{K}};\mathbb{Z})=H^6(\mathcal{Z}_{\mathcal{K}};\mathbb{Z})=0$.

Also we have

$$\begin{aligned}H^3(\mathcal{Z}_{\mathcal{K}};\mathbb{Z})&=\bigoplus_{J\subseteq\{1,\dots,5\}}\widetilde{H}^{2-|J|}(\mathcal{K}_{\mathcal{J}};\mathbb{Z})\\&=\widetilde{H}^0(\mathcal{K}_{\{1,3\}};\mathbb{Z})\oplus\widetilde{H}^0(\mathcal{K}_{\{1,4\}};\mathbb{Z})\oplus\widetilde{H}^0(\mathcal{K}_{\{2,4\}};\mathbb{Z})\oplus\widetilde{H}^0(\mathcal{K}_{\{2,5\}};\mathbb{Z})\oplus\widetilde{H}^0(\mathcal{K}_{\{3,5\}};\mathbb{Z})\\&=\mathbb{Z}[\alpha_{\{3\}}]\oplus\mathbb{Z}[\alpha_{\{4\}}]\oplus\mathbb{Z}[\alpha_{\{4\}}]\oplus\mathbb{Z}[\alpha_{\{5\}}]\oplus\mathbb{Z}[\alpha_{\{5\}}]\\&=\mathbb{Z}[u_1v_3]\oplus\mathbb{Z}[u_1v_4]\oplus\mathbb{Z}[u_2v_4]\oplus\mathbb{Z}[u_2v_5]\oplus\mathbb{Z}[u_3v_5]\end{aligned}$$

$$\begin{aligned}H^4(\mathcal{Z}_{\mathcal{K}};\mathbb{Z})&=\bigoplus_{J\subseteq\{1,\dots,5\}}\widetilde{H}^{3-|J|}(\mathcal{K}_{\mathcal{J}};\mathbb{Z})\\&=\widetilde{H}^0(\mathcal{K}_{\{1,2,4\}};\mathbb{Z})\oplus\widetilde{H}^0(\mathcal{K}_{\{2,3,5\}};\mathbb{Z})\oplus\widetilde{H}^0(\mathcal{K}_{\{1,3,4\}};\mathbb{Z})\oplus\widetilde{H}^0(\mathcal{K}_{\{2,4,5\}};\mathbb{Z})\oplus\widetilde{H}^0(\mathcal{K}_{\{1,3,5\}};\mathbb{Z})\\&=\mathbb{Z}[\alpha_{\{4\}}]\oplus\mathbb{Z}[\alpha_{\{5\}}]\oplus\mathbb{Z}[\alpha_{\{1\}}]\oplus\mathbb{Z}[\alpha_{\{2\}}]\oplus\mathbb{Z}[\alpha_{\{3\}}]\\&=\mathbb{Z}[-u_1u_2v_4]\oplus\mathbb{Z}[-u_2u_3v_5]\oplus\mathbb{Z}[-u_3u_4v_1]\oplus\mathbb{Z}[-u_4u_5v_2]\oplus\mathbb{Z}[u_1u_5v_3]\end{aligned}$$

and

$$\begin{aligned} H^7(\mathcal{Z}_{\mathcal{K}};\mathbb{Z}) &= \bigoplus_{J\subseteq\{1,\dots,5\}} \widetilde{H}^{6-|J|}(\mathcal{Z}_{\mathcal{K}};\mathbb{Z}) \\ &= \widetilde{H}^1(\mathcal{K};\mathbb{Z}) \\ &= \mathbb{Z}[\alpha_{\{1,2\}}] \\ &= \mathbb{Z}[-u_3u_4u_5v_1v_2] \end{aligned}$$

where wee identified the generators in Hochster's formula to generators in the DGA via the given formulas. Denote the generators of H^3 by $x_1,\dots,x_5$ and denote the generators of H^4 by $y_1,\dots,y_5$ and denote the generator of H^7 by z. Then we have the relations $x_1y_4 = x_2y_2 = -x_3y_5 = x_4y_3 = x_5y_1 = z$, and 0 otherwise. Thus relabelling the generators give

$$H^*(\mathcal{Z}_{\mathcal{K}};\mathbb{Z}) \cong \frac{\Lambda_{\mathbb{Z}}[x_1,\dots,x_5,y_1,\dots,y_5]}{(x_iy_i - x_jy_j, x_iy_j, x_ix_j, y_i, y_j \mid 1\leq i\neq j\leq 5)}$$

where $\deg(x_i) = 3$ and $\deg(y_i) = 4$. Notice that this is precisely the cohomology ring of $(S^3\times S^4)^{\#5}$.

There are three descriptions of the cohomology groups:

$$H^k(\mathcal{Z}_{\mathcal{K}};\mathbb{Z}) \cong \bigoplus_{2q-p=k} H_{-p}\left(\Lambda\left(\bigoplus_{i=1}^n \mathbb{Z}[\mathcal{K}]u_i\right)\right)_{2q} = \bigoplus_{J\subseteq\{1,\dots,n\}} \widetilde{H}^{k-|J|-1}(\mathcal{K}_J;\mathbb{Z})$$

For $L\subseteq J$ such that α_L is a cochain basis in the Hochster formula, we have

$$[\varepsilon(L,J)\chi(J\setminus L, L)] \leftrightarrow [\varepsilon(L,J)u_{J\setminus L}v_L] \leftrightarrow \alpha_L$$

Proposition 3.3.10
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots.n\}$. Then the following are true.

- $\text{rank}(H^0(C^{-*,0}(\mathcal{Z}_{\mathcal{K}};\mathbb{Z}))) = 1$.
- $\text{rank}(H^0(C^{-*,2q}(\mathcal{Z}_{\mathcal{K}};\mathbb{Z}))) = 0$ for $q\neq 0$.
- $\text{rank}(H^{-p}(C^{-*,2q}(\mathcal{Z}_{\mathcal{K}};\mathbb{Z})))$ for $p>q$ or $q>n$.
- $\text{rank}(H^{-p}(C^{-*,2q}(\mathcal{Z}_{\mathcal{K}};\mathbb{Z})))$ for $p\geq q>0$ or $q-p>\dim(\mathcal{K})+1$.
- $\text{rank}(H^{-1}(C^{-*,4}(\mathcal{Z}_{\mathcal{K}};\mathbb{Z}))) = \binom{n}{2} - f_1$ (recall that f_1 is the number of edges of $\mathcal{K}$).
- $\text{rank}(H^{-(n-\dim(\mathcal{K})+1)}(C^{-*,2n}(\mathcal{Z}_{\mathcal{K}};\mathbb{Z}))) = \text{rank}(\widetilde{H}^{\dim(\mathcal{K})}(\mathcal{K}))$.

Proposition 3.3.11
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots.n\}$. Then the following are true.

- $\text{rank}(H^0(\mathcal{Z}_{\mathcal{K}};\mathbb{Z})) = 1$.
- $\text{rank}(H^1(\mathcal{Z}_{\mathcal{K}};\mathbb{Z})) = \text{rank}(H^2(\mathcal{Z}_{\mathcal{K}};\mathbb{Z})) = 0$.
- $\text{rank}(H^3(\mathcal{Z}_{\mathcal{K}};\mathbb{Z})) = \binom{n}{2} - f_1$ (recall that f_1 is the number of edges of $\mathcal{K}$).
- $\text{rank}(H^{n+\dim(\mathcal{K})+1}(\mathcal{Z}_{\mathcal{K}};\mathbb{Z})) = \text{rank}(\widetilde{H}^{\dim(\mathcal{K})}(\mathcal{K}))$.

Lemma 3.3.12
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. If $\mathcal{K}$ is a triangulation of S^{m-1}, then the top cohomology $H^{n+m}(\mathcal{Z}_{\mathcal{K}};\mathbb{Z})$ is generated by $[\chi(\{1,\dots,n\}\setminus I, I)]$ for any $I\in\mathcal{K}$ and such that $|I| = n$.

If we use the Koszul complex as the cohomology model for $\mathcal{Z}_{\mathcal{K}}$, then the top cohomology is generated by $u_{\{1,\dots,n\}\setminus I}v_I$ for any $I\in\mathcal{K}$ such that $|I| = n$.

3.4 Geometry of Moment Angle Complexes

Definition 3.4.1 (Coordinate Subspaces)
Let $I \subseteq \{1, \dots, n\}$. Define the coordinate subspace of I to be

$$L_I = \{(z_1, \dots, z_n) \in \mathbb{C}^n \mid z_j = 0 \text{ for all } j \in I\}$$

Definition 3.4.2 (Coordinate Subspace Arrangement)
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Define the coordinate subspace arrangement of $\mathcal{K}$ to be

$$U(\mathcal{K}) = \mathbb{C}^n \setminus \bigcup_{\sigma \in \mathcal{K}} L_\sigma$$

Proposition 3.4.3
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then we have $U(\mathcal{K}) = (\mathbb{C}, \mathbb{C}^\times)^{\mathcal{K}}$.

Theorem 3.4.4
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. The map $(\mathbb{C}, \mathbb{C}^\times) \to (D^2, S^1)$ induces a T^n-equivariant deformation retract

$$U(\mathcal{K}) \simeq \mathcal{Z}_{\mathcal{K}}$$

3.5 Intersection of Quadrics

Let $P \subseteq \mathbb{R}^n$ be a polytope defined by

$$P = \{x \in \mathbb{R}^n \mid \langle a_i, x \rangle + b_i \geq 0 \text{ for } 1 \leq i \leq m\}$$

such that $a_1, \dots, a_m$ span $\mathbb{R}^n$. Let

$$A = \begin{pmatrix} a_1 \\ \vdots \\ a_m \end{pmatrix} \in M_{m \times n}(\mathbb{R}) \quad \text{and} \quad b = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix} \in \mathbb{R}^m$$

Recall that

$$P = \{x \in \mathbb{R}^n \mid Ax + b \geq 0\}$$

Definition 3.5.1 (Moment Angle Polytopes)
Let $P \subseteq \mathbb{R}^n$ be a polytope defined by $A \in M_{m \times n}(\mathbb{R})$ and $b \in \mathbb{R}^m$ such that $a_1, \dots, a_m$ span $\mathbb{R}^n$. Let $i_P : \mathbb{R}^n \to \mathbb{R}^m$ be the map $i_P(x) = Ax - b$. Let $\mu : \mathbb{C}^m \to \mathbb{R}^m$ be the map $\mu(z_1, \dots, z_m) = (|z_1|^2, \dots, |z_m|^2)$. Define the moment angle polytope $\mathcal{Z}_{A,b}$ to be the pullback

$$\begin{tikzcd} \mathcal{Z}_{A,b} \arrow[r] \arrow[d] & \mathbb{C}^n m \arrow[d, "\mu"] \\ P \arrow[r, hook, "i_P"] & \mathbb{R}^m \end{tikzcd}$$

Proposition 3.5.2
Let $P \subseteq \mathbb{R}^n$ be a polytope defined by $A \in M_{m \times n}(\mathbb{R})$ and $b \in \mathbb{R}^m$ such that $a_1, \dots, a_m$ span $\mathbb{R}^n$. Then there is a homeomorphism

$$\mathcal{Z}_{A,b} \cong \left\{ z \in \mathbb{C}^m \;\middle|\; \Gamma \begin{pmatrix} |z_1|^2 \\ \vdots \\ |z_m|^2 \end{pmatrix} = \Gamma b \right\}$$

where $\Gamma = \begin{pmatrix} \gamma_1 \\ \vdots \\ \gamma_{m-n} \end{pmatrix}$ and $\gamma_1, \dots, \gamma_{m-n}$ is a basis for $\ker(A^T)$.

This makes $Z_{A,b}$ an intersection of quadrics in $\mathbb{R}$, as well as an intersection of Hermitian quadrics (Hermitian since $|z_i|^2 = z_i\overline{z_i}$).

Proposition 3.5.3
Let $P \subseteq \mathbb{R}^n$ be a polytope defined by $A \in M_{m\times n}(\mathbb{R})$ and $b \in \mathbb{R}^m$ such that $a_1, \dots, a_m$ span $\mathbb{R}^n$. Then P is simple if and only if $\mathcal{Z}_{A,b}$ is a smooth manifold of real dimension $m+n$.

Smoothness criterion.

one to one between polytopes and intesections of quadrics

Proposition 3.5.4
Let $P \subseteq \mathbb{R}^n$ be a polytope defined by $A \in M_{m\times n}(\mathbb{R})$ and $b \in \mathbb{R}^m$ such that $a_1, \dots, a_m$ span $\mathbb{R}^n$. If P is simple, then there is a T^m-equivariant homeomorphism

$$\mathcal{Z}_{\mathcal{K}_P} \cong \mathcal{Z}_{A,b}$$

Theorem 3.5.5 (de Medrano)
Let $P \subseteq \mathbb{R}^n$ be a simple polytope. Suppose that $\mathcal{Z}_{\mathcal{K}_P}$ is a non-empty and non-degenerate intersection of 3 quadrics. Then exactly one of the following hold.

- P is combinatorially equivalent to a product of three simplicies, and there is a diffeomorphism
$$\mathcal{Z}_{\mathcal{K}_P} \cong S^{n_1} \times S^{n_2} \times S^{n_3}$$
where n_1, n_2, n_3 are odd.
- There is a diffeomorphism
$$\mathcal{Z}_{\mathcal{K}_P} \cong \#_{k=3}^{n+1}(S^k \times S^{2n+3-k})^{\# q_k}$$
for some q_k determined by the Gale diagram of P.

4 Real Moment Angle Complexes

Definition 4.0.1 (Real Moment Angle Complexes)
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Define the real moment angle complex of $\mathcal{K}$ to be

$$\mathbb{R}\mathcal{Z}_\mathcal{K} = (D^1, S^0)^\mathcal{K}$$

Proposition 4.0.2
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then the following diagram is a pullback

$$\begin{tikzcd}
\mathbb{R}\mathcal{Z}_\mathcal{K} \arrow[r] \arrow[d] & (D^1)^n \subseteq \mathbb{C}^n \arrow[d, "\mu"] \\
\text{cc}(\mathcal{K}) \arrow[r, hook] & I^n
\end{tikzcd}$$

where $D^1 \subseteq \mathbb{R}$ is the closed unit disc and $\mu : (D^2)^n \to I^n$ is the map given by $(x_1,\dots,x_n) \mapsto (x_1^2,\dots,x_n^2)$.

Example 4.0.3
We have

$$\mathcal{Z}_{\partial\Delta^{n-1}} \cong S^{n-1}$$

Proposition 4.0.4
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$ that is a triangulation of S^m. Then $\mathcal{Z}_\mathcal{K}$ is a compact topological manifold of dimension $m+1$.

Example 4.0.5
Let $\mathcal{K}$ be the boundary of an n-gon. Then there is a homeomorphism

$$\mathbb{R}\mathcal{Z}_\mathcal{K} \cong \Sigma_{1+(n-4)2^{n-3}}$$

Corollary 4.0.6
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then there is a homotopy equivalence

$$\Sigma\mathbb{R}\mathcal{Z}_\mathcal{K} \simeq \bigvee_{J\notin\mathcal{K}} \Sigma^2|\mathcal{K}_J|$$

4.1 The Fat Wedge Filtration of Real Moment Angle Complexes

Recall that cubical complex of a simplicial complex. We know that

$$(D^1,S^0)^\sigma = \bigcup_{\substack{\mu\subseteq\tau\subseteq\text{Vert}(\mathcal{K})\\ \tau\setminus\mu=\sigma}} C_{\mu\subseteq\tau}$$

and

$$\mathbb{R}\mathcal{Z}_\mathcal{K} = \bigcup_{\sigma\in\mathcal{K}} \bigcup_{\substack{\mu\subseteq\tau\subseteq\text{Vert}(\mathcal{K})\\ \tau\setminus\mu=\sigma}} C_{\mu\subseteq\tau}$$

We consider the fat wedge filtration of $\mathbb{R}\mathcal{Z}_\mathcal{K}$. We note that at the ith step, we have

$$\mathbb{R}\mathcal{Z}_\mathcal{K}^i = \bigcup_{I\subseteq\{1,\dots,n\},|I|=i} \mathbb{R}\mathcal{Z}_{\mathcal{K}_I}$$

Proposition 4.1.1
Let $n \in \mathbb{N}^+$. Let $\mathcal{K}$ be a simplicial complex on n. Let $J \subseteq \mathcal{K}$. Then we have

$$(|\text{Cone}(\text{Bd}(\mathcal{K}_J))|, |\text{Bd}(\mathcal{K}_J)|) \cong (\mathbb{R}\mathcal{Z}_{\mathcal{K}_J}, \mathbb{R}\mathcal{Z}_{\mathcal{K}_J}^{|J|-1})$$

Proof
Let $i_c : |\Delta^n| \to I^{n+1}$ denote the cubical embedding. For each $J \subseteq \{1, \dots, n\}$, the embedding restricts to a map $i_c^J : |\mathcal{K}_J| \to I^J$ where $I^J = (I, 0)^J$ is the moment angle complex of I. For each $\mu \subseteq \tau \subseteq J$, denote $C_{\mu\subseteq\tau}^J$ the $C_{\mu\subseteq\tau}$ face of I^J (so $C_{\mu\subseteq\tau}^J$ still lives inside I^J but some coordinates are 0). We compute that

$$\mathbb{R}\mathcal{Z}_{\mathcal{K}_J}^{|J|-1} = \bigcup_{A\subseteq J, |A|=|J|-1} \bigcup_{\sigma\in\mathcal{K}_A} \bigcup_{\substack{\mu\subseteq\tau\subseteq A \\ \tau\setminus\mu=\sigma}} C_{\sigma\subseteq\tau}^A = |\text{Bd}(\mathcal{K}_J)|$$

and similarly

$$\mathbb{R}\mathcal{Z}_{\mathcal{K}_J} = \bigcup_{\sigma\in\mathcal{K}_J} \bigcup_{\substack{\mu\subseteq\tau\subseteq J \\ \tau\setminus\mu=\sigma}} C_{\mu\subseteq\tau}^J = |\text{Cone}(\text{Bd}(\mathcal{K}_J))|$$

■

Let us verify this for the case $\mathcal{K} = \partial\Delta^2$. Pick $I = \{1\}$. Let $i_c : |\mathcal{K}| \to I^n$ be the map that is the cubical embedding. Now the relative version of the identity

$$(D^1, S^0)^\sigma = \bigcup_{\substack{\mu\subseteq\tau\subseteq\text{Vert}(\mathcal{K}) \\ \tau\setminus\mu=\sigma}} C_{\mu\subseteq\tau}$$

says that if $\sigma \in \mathcal{K}_I$, then

$$(D^1, S^0)^\sigma = \bigcup_{\substack{\mu\subseteq\tau\subseteq\text{Vert}(\mathcal{K}) \\ \tau\setminus\mu=\sigma}} C_{\mu\subseteq\tau}^I$$

where we consider $(D^1, S^0)^\sigma$ as a subspace of $\mathcal{Z}_\mathcal{K}$ by considering $\mathcal{Z}_{\mathcal{K}_I}$ as a subspace of $\mathcal{Z}_\mathcal{K}$, and on the right $C_{\mu\subseteq\tau}^I$ is the usual meaning but restricted to the cube $(D^1)^I$ of I^3.

Restricting this map to $\mathcal{K}_I$, this the map sending $\{1\}$ to $C_{\{1\},\{1\}}^I$ which is the point $-1 \in C_{\{1\},\{1\}}^I = D^1$, but is the point $(-1, 1, 1)$ in I^3.

5 Research Notes

The Foundation

lMoment angle complexes are T^n-equivariant objects defined using combinatorial data. More specifically, choose a simplicial complex $\mathcal{K}$ on $\{1, \dots, n\}$. For each simplex $\sigma \in \mathcal{K}$, define the space

$$(D^2, S^1)^\sigma = \prod_{i=1}^n Y_i$$

where

$$Y_i = \begin{cases} D^2 & \text{if } i \in \sigma \\ S^1 & \text{otherwise} \end{cases}$$

In other words, $(D^2, S^1)^\sigma$ is a product of D^2 and S^1 depending on how many vertices of $\mathcal{K}$ lie in σ. For any inclusion of simplicies $\sigma \subseteq \tau$, there is an inclusion map $(D^2, S^1)^\sigma \hookrightarrow (D^2, S^1)^\tau$. Then the definition of the moment angle complex $\mathcal{Z}_\mathcal{K}$ of $\mathcal{K}$ is just the gluing of products of D^2 and S^1 along these inclusions, or

$$\mathcal{Z}_\mathcal{K} = \bigcup_{\sigma \in \mathcal{K}} (D^2, S^1)^\sigma$$

Being combinatorial in nature, they form a testing ground for conjectures in topology. Moreover, in good cases (eg $\mathcal{K}$ is a triangulation of S^m), the moment angle complex $\mathcal{Z}_\mathcal{K}$ has the structure of a smooth manifold and they form a testing ground for conjectures in geometry.

These topological objects are also interesting because they lie at the intersection of a number of active fields. Other than moment angle complexes, there are other objects in topology and geometry that are defined in terms of combinatorial objects

$$\mathcal{Z}_\mathcal{K} \qquad\qquad \mathcal{Z}_P \qquad\qquad V_P$$

(Topology) (Symplectic Geometry) (Algebraic Geometry)

Moment angle polytopes $\mathcal{Z}_P$ are intersection of Hermitian quadrics that define non-Kahler complex manifolds, also makes for a testing ground for theories in symplectic and complex geometry.

Projective toric varieties V_P are combinatorial objects in algebraic geometry formed by certain types of polytopes that are also, explicit enough to form a testing ground for conjectures in algebraic geometry.

Subject to constraints, there is a relation between fans, polytopes and simplicial complexes

$$\begin{tikzcd}
& \text{Fans} \arrow[dl, shift left] \arrow[dr] & \\
\text{Simple Polytopes} \arrow[ur, shift left] \arrow[rr] & & \text{Simplicial Complexes}
\end{tikzcd}$$

Choosing our polytopes carefully connects the three objects into a unified theory.

$$\begin{tikzcd}
U(\mathcal{K}) \arrow[r, "\cong_{T^m}"] \arrow[d, "\simeq_{T^m}"] & U(\Sigma_P) \arrow[r, "T^m\text{-Orbit}"] \arrow[d, "\simeq_{T^m}"] & U(\Sigma_P)/T^m \arrow[r, "\cong"] \arrow[d, "\cong\text{diffeo}"] & V_P \\
\mathcal{Z}_\mathcal{K} \arrow[r, "\cong_{T^m}"] & \mathcal{Z}_P \arrow[r, "T^{m-n}\text{-Orbit}"] & \mathcal{Z}_P/T^{m-n} &
\end{tikzcd}$$

(Topology) (Symplectic Geometry) (Algebraic Geometry)

In fact, the first connection between symplectic geometry and combinatorics came from the following theorem.

Theorem 5.0.1 (The Atiyah–Guillemin–Sternberg convexity theorem (1982))
Let (M, ω) be a compact connected symplectic manifold. Let $T^k \times M \to M$ be a Hamiltonian torus action. Then $\mu(M)$ is a convex polytope in Lie$(T^k)^*$ given by

$$\mu(M) = \text{Conv}(\text{Fix}_{T^k}(M))$$

Delzant in 1988 proved that if the action is more over effective and M, then these polytopes completely determine the manifold structure. This motivates the definition of quasitoric manifolds: Smooth manifolds M^{2n} with a torus action T^n that locally looks like the standard action, together with a homeomorphism of the orbit space $M/T^n \cong P$ with a polytope. In 2008, the following conjecture is made.

Conjecture 5.0.2 (Masuda–Suh (2008))
If M and N are quasitoric manifolds such that

$$H^*(M) \cong H^*(N)$$

Then M and N are homeomorphic (or diffeomorphic).

We are interested in the cohomological rigidty of moment angle complexes instead.

The Focus of My Work

The first observation is that Hochster's formula shows that the cohomology of a moment angle complex is completely determined by the simplicial complex:

Theorem 5.0.3 (Hochster's Formula)
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then there is an isomorphism of rings

$$H^*(\mathcal{Z}_{\mathcal{K}}) \cong \bigoplus_{J \subseteq \{1,\dots,n\}} H^*(\mathcal{K}_J)$$

where
- $\mathcal{K}_J$ is the full subcomplex of $\mathcal{K}$ with the vertices in J.
- The additive isomorphism is given by

$$H^n(\mathcal{Z}_{\mathcal{K}}) \cong \bigoplus_{J \subseteq \{1,\dots,n\}} H^{n-|J|-1}(\mathcal{K}_J)$$

- The ring structure is given the map $H^i(\mathcal{K}_I) \otimes H^j(\mathcal{K}_J) \to H^{i+j+1}(\mathcal{K}_{I \cup H_J})$ which sends products of generators to generators if $I \cap J = \emptyset$ and 0 otherwise.

The second observation is that certain classes of simplicial complexes have the geometry of connected sums of products of spheres:

Theorem 5.0.4
Let P be a stacked polytope. Let $n = \dim(P)$ and $m = |V(P)|$. Then there is a diffeomorphism

$$\mathcal{Z}_{\partial P} \cong \#_{k=3}^{m-n+1} (S^k \times S^{m+n-k})^{(k-2)\binom{m-n}{k-1}}$$

Among the class of topological spaces, spheres are homologically and homotopically the easiest to understand. It is natural to ask the question of in what conditions does a moment angle complex has its cohomology only involving spheres, and in this case is it possible to lift it to a homotopy equivalence, a homeomorphism or a diffeomorphism. Currently my work focuses on moment angle complexes coming from simplicial polytopes of dimension 4 with 8 vertices.

What is Known in the Simplicial 4-Polytopes and 8-Vertices Cases

There are 39 simplicial complexes in dimension 4 with 8 vertices (no ghost vertices). Of which 37 of them come from the boundary of a simplicial 4-polytope with 8 vertices (polytopal spheres). These are classified in [An Enumeration of Simplicial 4-Polytopes with 8-Vertices] by Grunbaum and Sreedharan, and whose list was later corrected by my supervisor.

P_1^8 is a stacked polytope where:

- The starting simplex is 12356.
- Vertex 4 is stacked on the facet 1235.
- Vertex 7 is stacked on the facet 1236.
- Vertex 8 is stacked on the facet 1256.

P_2^8 is a stacked polytope where:

- The starting simplex is 12356.
- Vertex 4 is stacked on the facet 1235.
- Vertex 7 is stacked on the facet 1236.
- Vertex 8 is stacked on the facet 1245.

P_3^8 is a stacked polytope where:

- The starting simplex is 12345.
- Vertex 6 is stacked on the facet 1235.
- Vertex 7 is stacked on the facet 1236.
- Vertex 8 is stacked on the facet 1234.

For a stacked polytope P, we have a theorem

Theorem 5.0.5
Let P be a stacked polytope. Let $n = \dim(P)$ and $m = |V(P)|$. Then there is a diffeomorphism

$$\mathcal{Z}_{\partial P} \cong \#_{k=3}^{m-n+1}(S^k \times S^{m+n-k})^{(k-2)\binom{m-n}{k-1}}$$

Hence we conclude that

$$\mathcal{Z}_{\partial P_1^8} \cong \mathcal{Z}_{\partial P_2^8} \cong \mathcal{Z}_{\partial P_3^8} \cong (S^3 \times S^9)^{\#6} \# (S^4 \times S^8)^{\#8} \# (S^5 \times S^7)^{\#3}$$

Also P_{35}^8, P_{36}^8 and P_{37}^8 are neighbourly polytopes (one such is cyclic), and we have the following theorem.

Theorem 5.0.6 (Membrillo Solis, Theriault (2026))
Let $\mathcal{K}$ be a simplicial complex such that the following are true.

- $\mathcal{K}$ is a triangulation of S^{2n+1}.
- $\mathcal{K}$ is neighbourly.

Then there is a homotopy equivalence

$$(D^k, S^{k-1})^{\mathcal{K}} \simeq \#_{i=1}^{j} \left(S^{t_i} \times S^{s_i}\right)$$

for some $j \in \mathbb{N}^+$ and some $t_i, s_i \in \mathbb{N}^+$ for all $k \geq 2$.

So we have

$$\mathcal{Z}_{\partial P_{35}^8} \simeq \mathcal{Z}_{\partial P_{36}^8} \simeq \mathcal{Z}_{\partial P_{37}^8} \simeq (S^5 \times S^7)^{\#16} \# (S^6 \times S^6)^{\#15}$$

Also P_{34}^8 is a cross polytope (It is given by $P_{34}^8 = S^0 * S^0 * S^0 * S^0$).

Theorem 5.0.7 (Fan, Cheng, Ma, Wang (2014))
Let $\mathcal{K}$ be a simplicial complex such that $\mathcal{K}$ is a triangulation of S^n for $n \geq 2$. Suppose that

$$H^*(\mathcal{Z}_{\mathcal{K}}) \cong H^*\left(\#_{i=1}^j\left(\prod_{k=1}^{t_i} S^{d_k}\right)\right)$$

Then the following are true.

- If $t_i = n+1$ for some i, then $j = 1$ and there is a homeomorphism
$$\mathcal{Z}_{\mathcal{K}} \cong \prod_{i=1}^{n+1} S^3$$
- We have
$$\left|\left\{i \in \mathbb{N} \;\middle|\; t_i \geq \left\lfloor \frac{n}{2} \right\rfloor + 2\right\}\right| \leq 1$$

(MOMENT-ANGLE MANIFOLDS AND CONNECTED SUMS OF SPHERE PRODUCTS)

Either by direction computation or the following theorem, we see that

$$H^*(\mathcal{Z}_{\partial P_{34}^8}) \cong H^*(S^3 \times S^3 \times S^3 \times S^3)$$

and it lifts to a homeomorphism
$$\mathcal{Z}_{\partial P_{34}^8} \cong S^3 \times S^3 \times S^3 \times S^3$$

Theorem 5.0.8 ((Oganisian, Panov (2024)))
Let $\mathcal{K}$ be a simplicial complex such that $\mathcal{K}$ is a triangulation of S^3. Then

$$H^*(\mathcal{Z}_{\mathcal{K}}) \cong H^*\left(\#_{i=1}^j\left(\prod_{k=1}^{t_i} S^{d_k}\right)\right)$$

if and only if exact;y one of the following are true.

- $\mathcal{K} = S^0 * S^0 * S^0 * S^0$.
- $G(\mathcal{K}^1)$ is chordal.
- $\mathcal{K}^1$ has exactly two missing edges with distinct vertices (forming a square).

The same paper by Fan, Cheng, Ma, Wang also shows:

$$H^*(\mathcal{Z}_{\partial P_{27}^8}) \cong H^*(\mathcal{Z}_{\partial P_{28}^8}) \cong H^*(\mathcal{Z}_{\partial P_{29}^8}) \cong H^*\left((S^3 \times S^3 \times S^6)\#(S^5 \times S^7)^{\#8}\#(S^6 \times S^6)^{\#8}\right)$$

The paper [Iriye (2016)] ON THE MOMENT-ANGLE MANIFOLD CONSTRUCTED BY FAN, CHEN, MA AND WANG proves that there is a diffeomorphism

$$\mathcal{Z}_{\partial P_{28}^8} \cong (S^3 \times S^3 \times S^6)\#(S^5 \times S^7)^{\#8}\#(S^6 \times S^6)^{\#8}$$

Summary:

- The diffeomorphism types of the moment angle complexes of $P_1^8, P_2^8, P_3^8, P_{28}^8$ are understood (they are the stacked polytopes of dim 4 and vert 8 together with the example given by FCMW).
- The homeomorphism type of the moment angle complexe of P_{34}^8 is understood (it is the cross polytope of dim 4).
- The homotopy types of moment angle complexes of $P_{35}^8, P_{36}^8, P_{37}^8$ are understood (they are the neighbourly polytopes of dim 4 and vert 8).
- The conditions for the cohomology type of moment angle complexes of simplicial complexes of dimension 3 (and hence simplicial 4-polytopes) to be the connected sum of products of spheres is understood.

Techniques and Research Directions

Going forward, there are a number of possible directions that I will focus my work on, in terms of the cohomology, the topology and the rigidity of moment angle complexes.

Analyzing the Cohomology

If $\mathcal{K}$ is a simplicial complex that is a triangulation of S^n. Then the following are true.

- $\mathcal{K}_{[m]\setminus J}$ is a deformation retract of $\mathcal{K}\setminus\mathcal{K}_J$.
- $|\mathcal{K}_J|$ is a compact locally contractible subspace of $|\mathcal{K}|$.
- By Alexander duality, we have

$$\widetilde{H}^i(\mathcal{K}_J)\cong\widetilde{H}_{n-i-1}(S^n\setminus\mathcal{K}_J)\cong\widetilde{H}_{n-i-1}(\mathcal{K}\setminus\mathcal{K}_J)\cong\widetilde{H}_{n-i-1}(\mathcal{K}_{[m]\setminus J})$$

I have written python programs to compute the missing faces of a moment angle complex, which have significantly reduced the time needed to compute the ring structure in cohomology. I have also developed a simple graphing tool that visualizes subgraphs of the underlying graph of polytopes. Together with the equivalent characterization in [Oganisian, Panov (2024)], I found that only P_7^8, P_{16}^8, P_{17}^8, P_{22}^8 and P_{26}^8 do not have the cohomology of a connected sum of products of spheres, and all the others do.

I also explicitly computed

$$H^*(\mathcal{Z}_{\partial P_7^8})=H^*((S^3\times S^9)^{\#3}\#(S^4\times S^8)^{\#3}\#(S^5\times S^7)\#X)$$

where

$$H^*(X)=\frac{\mathbb{Z}\langle a_1,a_2,b_1,b_2,c,d_1,d_2,e,f_1,f_2,g_1,g_2,h\rangle}{I}$$

and I is the ideal generated by the following relations:

- $a_1a_2=d_2$.
- $a_ib_i=e$.
- $a_ic=f_i$.
- $a_1d_1=g_2,a_2d_2=g_1$.
- $b_ic=g_i$.
- Poincare relations $a_ig_i=b_if_i=ce=d_1d_2=h$.
- $z^2=0$ for all generators z.

and $\deg(a_i)=3$, $\deg(b_i)=4$, $\deg(c)=5$, $\deg(d_i)=6$, $\deg(e)=7$, $\deg(f_i)=8$, $\deg(h)=12$ (13 generators and 28 relations). It is interesting to find a model for X and to see find the homotopy type of $\mathcal{Z}_\mathcal{K}$ in these odd cases, and I aim to use Macaulay2 to simply the above expression and the cohomology rings of the other odd cases.

Analyzing the Topology

There are retractions $\mathcal{Z}_{\mathcal{K}_J}\hookrightarrow\mathcal{Z}_\mathcal{K}\to\mathcal{Z}_{\mathcal{K}_J}$ where $J\subseteq\{1,\dots,n\}$ induced by retracting (D^2,S^1) to (S^1,S^1) by mapping D^2 to the base point, and these are typically easier to compute. In particular, the retraction may factor through the $(n-1)$-skeleton, which may reveal information about the topology of the moment angle complex. In [Membrilo Solis, Theriault (2026)] and [Iriye (2018)], certain $\mathcal{Z}_{\mathcal{K}_J}$ are wedges of spheres which are useful to knowing the topology because the remainder of $\mathcal{Z}_{\mathcal{K}_J}$ in $\mathcal{Z}_\mathcal{K}$ may be understood by attaching maps of spheres to wedges of spheres, and if this is the case we may use Hilton Milnor theorem to analyze the components.

In the general case, [BBCG] only decomposes the suspension of $\mathcal{Z}_\mathcal{K}$ as a wedge of spheres:

$$\Sigma\mathcal{Z}_\mathcal{K}\simeq\bigvee_{J\notin\mathcal{K}}\Sigma^{2+|J|}|\mathcal{K}_J|$$

And work was done to desuspend this decomposition [Iriye, Kishimoto], the case for a shifted complex or disjoint points.

To understand $\mathcal{Z}_{\mathcal{K}}$ it is also useful to analyze fibrations related to $\mathcal{Z}_{\mathcal{K}}$. The following are some classical fiber bundles:

- $T^{m-n} \to \mathcal{Z}_{\mathcal{K}} \to M(P, \Lambda)$ to quasi-toric manifolds.
- $\mathcal{Z}_{\mathcal{K}} \to DJ(\mathcal{K}) = (\mathbb{CP}^\infty, *)^{\mathcal{K}} \to BT^m$.

Obstructions to Rigidity

Massey products cannot be detected in cohomological isomorphisms so it is a prime method to showing two spaces are not homeomorphic. There are combinatorial descriptions [Denham, Suciu (2005)] for whitehead products to be non-trivial using the underlying graph of simplicial complexes. More generally the same can be said for A_∞ structures on the cohomology ring.

6 Basic Definitions

6.1 Symplectic Manifolds

Definition 6.1.1 (Symplectic Manifolds)
Let M be a smooth manifold. Let $\omega \in \Omega^2(M)$ be a 2-form. We say that ω is a symplectic form if the following are true.

- ω is a symplectic structure on TM.
- $d\omega = 0$.

In this case we call (M, ω) a symplectic manifold.

Lemma 6.1.2
Let (M, ω) be a symplectic manifold. Then $\dim(M)$ is even.

Definition 6.1.3 (The Liouville Volume Form)
Let (M, ω) be a symplectic manifold of dimension $2n$. Define the Liouville volume form of M by

$$\mu_M = \frac{\omega^n}{n!}$$

Proposition 6.1.4
Let (M, ω) be a symplectic manifold. Then the following are true.

- μ_M is a nowhere-vanishing differential form.
- M is orientable.

Example 6.1.5
The following are symplectic manifolds.

- $\mathbb{R}^{2n}$ with coordinates $(x_1, \dots, x_n, y_1, \dots, y_n)$ and with symplectic form

$$\sum_{i=1}^n dx_i \wedge dy_i$$

- For all $g \geq 1$, the compact orientable surface Σ_g with any ω such that $0 \neq [\omega] \in H^2_{\text{dR}}(\Sigma_g)$.

6.2 Standard Symplectic Coordinates

Theorem 6.2.1 (Moser's Theorem Relative Version)
Let M be a smooth manifold. Let X be an embedded compact submanifold of M. Let $\iota : X \hookrightarrow M$ be the inclusion map. Let ω_0 and ω_1 be symplectic forms on M. If $\omega_0|_X = \omega_1|_X$, then there exists $X \subseteq U_0, U_1 \subseteq M$ and a map $\varphi : U_0 \to U_1$ such that the following are true.

- $\iota = \varphi \circ \iota$.
- φ is a diffeomorphism.
- $(\varphi|_X)^*\omega_1 = \omega_0$.

Theorem 6.2.2 (Darboux (1982))
Let (M, ω) be a symplectic manifold. Let $p \in M$. Then there exists a chart $(U, x^1, \dots, x^n, y^1, \dots, y^n)$ at p such that

$$\omega|_U = \sum_{i=1}^n dx^i \wedge dy^i$$

Definition 6.2.3 (Darboux Coordinates)
Let (M, ω) be a symplectic manifold. Let $(U, x^1, \dots, x^n, y^1, \dots, y^n)$ be a local chart of M. We say

that the chart is Darboux if

$$\omega|_U = \sum_{i=1}^n dx^i \wedge dy^i$$

6.3 Maps Preserving the Symplectic Structure

Definition 6.3.1 (Symplectic Maps)
Let (M, ω) and (N, σ) be symplectic manifolds. Let $f : M \to N$ be a smooth map.

- We say that f is symplectic if
$$f^*\sigma = \omega$$
- We say that f is a symplectomorphism if f is symplectic and a diffeomorphism.

Lemma 6.3.2
Let (M, ω) and (N, σ) be symplectic manifolds. Let $f : M \to N$ be a symplectic map. Then f is an immersion.

Definition 6.3.3 (The Group of Symplectomorphisms)
Let (M, ω) be a symplectic manifold. Define the group of symplectomorphisms of M to be the subgroup

$$\text{Symp}(M, \omega) = \{f : M \to M \mid f \text{ is a symplectomorphism }\} \leq \text{Diff}(M)$$

6.4 Symplectic Structure on the Cotangent Bundle

Through out this section we denote the projection map $T^*M \to M$ by π_M.

Definition 6.4.1 (The Canonical Hamiltonian Structure)
Let M be a smooth manifold. Define the canonical Hamiltonian structure ω on T^*M as follows. At the local chart $(U, q^1, \dots, q^n, p^1, \dots, p^n)$, ω is given by

$$\omega|_U = \sum_{i=1}^n dq^i \wedge dp^i$$

Lemma 6.4.2
Let M be a smooth manifold. Let ω be the canonical Hamiltonian structure. Then (T^*M, ω) is a symplectic manifold.

Definition 6.4.3 (The Liouville 1-Form)
Let M be a smooth manifold. Let $\pi : T^*M \to M$ be the bundle map. Define the Lioville 1-form $\theta \in \Omega^1(T^*M)$ as follows. For each $q \in M$ and $p \in T_q^*M$, define

$$\theta_{(q,p)} = p(\pi_M(v))$$

for any $v \in T_{(q,p)}(T^*M)$.

Proposition 6.4.4
Let M be a smooth manifold. Let $(U, q^1, \dots, q^n, p^1, \dots, p^n)$ be a chart of T^*M where $(q_1, \dots, q_n) \in M$. Then θ is given in local coordinates by

$$\theta|_U = \sum_{i=1}^n p^i dq^i$$

Proposition 6.4.5
Let M be a smooth manifold. Let θ be the Liouville 1-form. Let ω be the canonical Hamitonian structure. Then we have
$$\omega = -d\theta$$

Definition 6.4.6 (The Lift of a Diffeomorphism)
Let M, N be smooth manifolds. Let $f : M \to N$ be a diffeomorphism. Define the lift of f to be the map $f_\# : T^*M \to T^*N$ given by
$$f_\#(p, v) = (f(p), w)$$
where $w \in T_p^*M$ is the unique element such that $v = (df_p)^*w$.

Proposition 6.4.7
Let M, N be smooth manifolds. Let $f : M \to N$ be a diffeomorphism. Then the following are true.
- We have $f \circ \pi_M = \pi_N \circ f_\#$.
- $f_\#^*\theta_N = \theta_M$.

7 Lagrangian Submanifolds

Definition 7.0.1 (Lagrangian Vector Subspaces)
Let k be a field. Let (V, ω) be a symplectic vector space over k. Let $W \subseteq V$ be a subspace. We say that W is Lagrangian if the following are true.

- $\omega|_W = 0$.
- $2\dim(W) = \dim(V)$.

Definition 7.0.2 (Lagrangian Submanifolds)
Let (M, ω) be a symplectic manifold. Let L be an embedded submanifold of M. We say that L is Lagrangian if T_pL is a Lagrangian subspace of M.

Lemma 7.0.3
Let (M, ω) be a symplectic manifold. Let L be an embedded submanifold of M. Then L is Lagrangian if and only if the following are true.

- $\iota^*\omega = 0$ where $\iota : L \hookrightarrow M$ is the inclusion map.
- $2\dim(L) = \dim(M)$.

Proposition 7.0.4
Let M be a smooth manifold. Let $s \in \Omega^1(M)$. Then $\text{im}(s)$ is a Lagrangian submanifold of T^*M if and only if $ds = 0$.

Theorem 7.0.5 (Weinstein Lagrangian Neighborhood Theorem)
Let M be a smooth manifold. Let X be an embedded compact submanifold of M. Let $\iota : X \hookrightarrow M$ be the inclusion map. Let ω_0 and ω_1 be symplectic forms on M such that X is a Lagrangian submanifold of (M, ω_0) and (M, ω_1). Then there exists $X \subseteq U_0, U_1 \subseteq M$ and a map $\varphi : U_0 \to U_1$ such that the following are true.

- $\iota = \varphi \circ \iota$.
- φ is a diffeomorphism.
- $(\varphi|_X)^*\omega_1 = \omega_0$.

7.1 Relation to Symplectomorphisms

Proposition 7.1.1
Let (M, ω) and (N, σ) be symplectic manifolds. Let $f : M \to N$ be a diffeomorphism. Then f is a symplectomorphism if and only if Γ_f is a Lagrangian submanifold of $(M \times N, \text{proj}_M^*\omega - \text{proj}_N^*\sigma)$.

Recipe for producing symplectomorphisms.

1. Start with a Lagrangian submanifold of $(T^*M \times T^*N, \omega \times \sigma)$.
2. Send the Lagrangian through the twist map $\tau : T^*M \times T^*N \to T^*M \times T^*N$ given by $(x, v) \mapsto (x, -v)$. The image is then a Lagrangian.
3. Check whether $\tau(\sigma)$ is the graph of some diffeomorphism $\varphi : M \to N$.
4. If it is, then φ is a symplectomorphism.

7.2 Generating Functions

Definition 7.2.1 (Generating Functions)
Let M, N be smooth manifolds. Let $f \in \mathcal{C}^\infty(M \times N)$. Define

$$Y_f = \{(p, q, \text{proj}_{T_p^* M} d_{(p,q)} f, -\text{proj}_{T_q^* M} d_{(p,q)} f) \in M \times N \times T^* M \times T^* N \mid (p, q) \in M \times N\}$$

We say that f is a generating function if there exists a diffeomorphism $\varphi : T^* M \to T^* N$ such that $Y_f = \Gamma_\varphi$.

In this case, we know that $\varphi : T^* M \to T^* N$ is given by

$$\varphi(x, \text{proj}_{T_p^* M} d_{(p,q)} f) = (y, -\text{proj}_{T_q^* M} d_{(p,q)} f)$$

8 Calculus on Symplectic Manifolds

8.1 Symplectic Vector Fields

Definition 8.1.1 (Symplectic Vector Fields)
Let (M, ω) be a symplectic manifold. Let $X \in \mathfrak{X}(M)$ be a vector field. We say that X is symplectic if $\iota_X(\omega)$ is a closed form.

Proposition 8.1.2
Let (M, ω) be a symplectic manifold. Let $X \in \mathfrak{X}(M)$ be a vector field. Let θ^X be the flow of X. Then X is symplectic if and only if $\theta^X(t, -)$ is a symplectomorphism for all t.

8.2 Hamiltonian Vector Fields

Definition 8.2.1 (The Vector Field of a Function)
Let (M, ω) be a symplectic manifold. Let $H \in \mathcal{C}^\infty(M)$. Define the associated vector field of H to be the unique vector field $X_H \in \mathfrak{X}(M)$ such that

$$\iota_{X_H}(\omega) = df$$

where ι is the interior product.

Proposition 8.2.2
Let (M, ω) be a symplectic manifold. Let $H \in \mathcal{C}^\infty(M)$. Let $(U, x^1, \dots, x^n, y^1, \dots, y^n)$ be a Darboux chart. Then the vector field of H is given locally at the chart by

$$(X_H)|_U = \sum_{i=1}^n \left(\frac{\partial H}{\partial y^i} \frac{\partial}{\partial x^i} - \frac{\partial H}{\partial x^i} \frac{\partial}{\partial y^i} \right)$$

Definition 8.2.3 (Hamiltonian Vector Fields)
Let (M, ω) be a symplectic manifold. Let $X \in \mathfrak{X}(M)$ be a vector field. We say that X is Hamiltonian if there exists a smooth map $H \in \mathcal{C}^\infty(M)$ such that

$$X = X_H$$

Proposition 8.2.4
Let (M, ω) be a symplectic manifold. Let $X \in \mathfrak{X}(M)$ be a vector field. If X is Hamiltonian, then X is symplectic.

Theorem 8.2.5 (Liouville's Theorem)
Let (M, ω) be a symplectic manifold. Let $X \in \mathfrak{X}(M)$ be a complete Hamiltonian vector field. Let θ^X be the flow of X. Then we have

$$\theta^X(t, -)^* \mu_M = \mu_M$$

8.3 The Poisson Structure

Definition 8.3.1 (The Poisson Bracket)
Let (M, ω) be a symplectic manifold. Define the poisson bracket of M to be the map $\{-, -\} : \mathcal{C}^\infty(M) \times \mathcal{C}^\infty(M) \to \mathcal{C}^\infty(M)$ by

$$\{f, g\} = X_f(g)$$

Lemma 8.3.2
Let (M, ω) be a symplectic manifold. Then the following are true.

- $(M, \{-,-\})$ is a Lie algebra.
- Leibniz rule: $\{f, gh\} = \{f, g\}h + g\{f, h\}$.

Proposition 8.3.3
Let (M, ω) be a symplectic manifold. Then the map $\mathcal{C}^\infty(M) \to \mathfrak{X}(M)$ given by $f \mapsto X_f$ is a Lie algebra homomorphism.

8.4 Integrable Systems

Definition 8.4.1 (Hamiltonian Systems)
A Hamiltonian system is a triple (M, ω, H) where (M, ω) is a symplectic manifold and $H \in \mathcal{C}^\infty(M)$.

Definition 8.4.2 (Integrable Systems)
Let (M, ω, H) be a Hamiltonian system. Suppose that $\dim(M) = 2n$. We say that it is integrable if there exists $f_i \in \mathcal{C}^\infty(M)$ for $1 \leq i \leq n$ and $f_1 = H$ such that

$$\{f_i, f_j\} = 0$$

for all i, j.

Theorem 8.4.3 (Arnold-Liouville Theorem)
Let $(M, \omega, H, f_2, \dots, f_n)$ be an integrable system. Let $c \in \mathbb{R}$. Then $(H, f_2, \dots, f_n)^{-1}(c)$ is a Lagrangian submanifold of M. Moreover, if M is compact and connected, then the following are true.

- $(H, f_2, \dots, f_n)^{-1}(c)$ is diffeomorphic to T^k for some $k \in \mathbb{N}^+$.
- Angle coordinates.

9 Symplectic and Hamiltonian Actions

9.1 Symplectic Actions

9.2 Moment Maps

Definition 9.2.1 (Moment Maps)
Let (M,ω) be a symplectic manifold. Let G be a Lie group acting on M by symplectomorphisms. A moment map of M is a map

$$\mu : M \to \text{Lie}(G)^* \cong \mathbb{R}^{\dim(G)}$$

such that the following are true.

- Hamilton's condition: For all $\xi \in \text{Lie}(G)$, the fundamental vector field ξ_M is Hamiltonian via the map $\langle \mu, \xi \rangle : M \to \mathbb{R}$.
- Equivariance: For all $g \in G$, we have

$$\mu(g \cdot x) = \text{Ad}_g^*(\mu(x))$$

  for all $x \in M$.

9.3 Hamiltonian Actions

9.4 Hamiltonian Torus Actions

Definition 9.4.1 (Hamiltonian Torus Actions)
Let (M,ω) be a symplectic manifold. Let T^k be a torus that acts on M by symplectomorphisms. We say that the action is Hamiltonian if it admits a moment map

$$\mu : M \to \text{Lie}(T^k)^*$$

Theorem 9.4.2 (The Atiyah–Guillemin–Sternberg convexity theorem (1982))
Let (M,ω) be a compact connected symplectic manifold. Let $T^k \times M \to M$ be a Hamiltonian torus action. Then $\mu(M)$ is a convex polytope in $\text{Lie}(T^k)^*$ given by

$$\mu(M) = \text{Conv}(\text{Fix}_{T^k}(M))$$

Definition 9.4.3 (Toric Symplectic Manifold)
Let (M,ω) be a symplectic manifold of dimension $2n$. Let $T^n \times M \to M$ be a Hamiltonian torus action. We say that M is a toric symplectic manifold if M is compact connected and the action is faithful.

Delzant polytope: simple, rational, smooth

Theorem 9.4.4 (Delzant (1988))
There is a one to one bijection

$$\frac{\{(M,\omega) \mid (M,\omega) \text{ is a toric symplectic manifold }\}}{\cong} \overset{1:1}{\leftrightarrow} \{P \mid P \text{ is a Delzant polytope }\}$$

10 Quasitoric Manifolds

10.1 The Davis-Januszkiewicz Classification Theorem

Definition 10.1.1 (Quasitoric Manifolds)
Let M be a smooth manifold of dimension $2n$. Let T^n be a torus acting smoothly on M. We say that M is a quasitoric manifold if the following are true.

- For each $p \in M$, there is a T^n-equivariant chart $(U, \varphi : U \to \mathbb{C}^n)$ where $\mathbb{C}^n$ is equipped with the standard torus action.
- There exists a convex polytope P such that there is a homeomorphism

$$M/T^n \cong P$$

Toric symplectic manifolds are quasitoric manifolds. Moment angle manifolds are NOT quasitoric manifolds.

Definition 10.1.2 (Orbit Type Strata)
Let M be a quasitoric manifold. Let P be the underlying convex polytope. Let $H \leq T^n$ be a subgroup. Define the orbit type strata of H to be

$$M_{(H)} = \{x \in M \mid \text{Stab}_{T^n}(x) = H\}$$

Proposition 10.1.3
Let M be a quasitoric manifold. Let P be the underlying convex polytope. Then there is a bijection

$$\{F \in \mathcal{F}(P) \mid F \text{ has codimension } k\} \overset{1:1}{\leftrightarrow} \left\{ N \subseteq M \;\middle|\; \begin{array}{c} N \text{ is an orbit type strata of subtori of} \\ T^n \text{ of dimension } k \end{array} \right\}$$

given as follows. Let $\pi : M \to P$ be the quotient map. Then we have

$$F_i \mapsto \pi^{-1}(\text{relint}(F))$$

Definition 10.1.4 (Primitive Vectors)
Let $v = (v_1, \dots, v_n) \in \mathbb{Z}^n$. We say that v is primitive if $v \neq \lambda u$ for any $\lambda \in \mathbb{Z} \setminus \{0\}$ and $u \in \mathbb{Z}^n$.

Definition 10.1.5 (The Characteristic Matrix)
Let M be a quasitoric manifold. Let P be the underlying convex polytope. Define the characteristic matrix Λ as follows. For each facet F_i of P for $1 \leq i \leq m$, the facet corresponds to a a circle $S_i^1 \subseteq T^n$ by the above. Every circle $S_i^1 \subseteq T^n$ corresponds to a unique primitive vector $v_i = (v_{1i}, \dots, v_{1n}) \in \mathbb{Z}^n$ up to sign. Then

$$\Lambda = \begin{pmatrix} v_{11} & \cdots & v_{1m} \\ \vdots & \ddots & \vdots \\ v_{n1} & \cdots & v_{nm} \end{pmatrix} \in M_{n \times m}(\mathbb{Z})$$

Definition 10.1.6 (Characteristic Pairs)
Let P be a simple polytope of dimension n. Let $F_1, \dots, F_m$ be the facets of P. Let $\Lambda \in M_{n \times m}(\mathbb{Z})$ be a matrix. Let $c_1, \dots, c_m$ be the columns of Λ. We say that (P, Λ) is a characteristic pair if the following are true.

- c_i is a primitive vector for all $1 \leq i \leq m$.
- If $F_{i_1} \cap \cdots \cap F_{i_n}$ is a vertex of P, then

$$\det \begin{pmatrix} | & & | \\ c_{i_1} & \cdots & c_{i_n} \\ | & & | \end{pmatrix} = \pm 1$$

Definition 10.1.7 (The Associated Quasitoric Manifold)
Let (P,Λ) be a characteristic pair. Suppose that $\dim(P)=n$. Let $F_1,\dots,F_m$ be the facets of P. Let $c_1,\dots,c_m$ be the columns of Λ. Define the associated quasi-toric manifold of (P,Λ) by

$$M(P,\Lambda)=\frac{P\times T^n}{\sim}$$

where $(p,t_1)\sim(p,t_2)$ if $t_2^{-1}t_1\in\langle c_i\mid p\in F(p)\rangle\cong T^k\leq T^n$ and $F(p)$ is the unique facet such that $p\in\text{relint}(F(p))$ for any $p\in P$.

Theorem 10.1.8 (Davis-Januszkiewicz Classification Theorem)
There is a one to one bijection

$$\frac{\{M\mid M\text{ is a quasitoric manifold}\}}{\cong}\overset{1:1}{\longleftrightarrow}\frac{\{(P,\Lambda)\text{ is a characteristic pair}\}}{\sim}$$

where $(P,\Lambda)\sim(Q,\Gamma)$ if P and Q are combinatorially equivalent and there exists $A\in GL(n,\mathbb{Z})$ with $\det(A)=\pm1$ such that $A\Lambda$ and Q are equal up to permutation of columns.

10.2 Basic Structures

Proposition 10.2.1
Let M be a quasitoric manifold with characteristic pair (P,Λ). Let $F_1,\dots,F_m$ be the facets of P. Then there is an isomorphism of graded rings

$$H^*(M;\mathbb{Z})\cong\frac{\mathbb{Z}[x_1,\dots,x_m]}{I_{\text{SR}}+I_\Lambda}$$

where
- $\deg(x_i)=2$ for $1\leq i\leq m$.
- I_{SR} is the Stanley-Reisner Ideal given by

$$I_{\text{SR}}=(xj_1\cdots x_{j_t}\mid F_{j_1}\cap\cdots\cap F_{j_t}=\emptyset)$$

- I_Λ is the linear relations given by

$$I_\Lambda=(\Lambda(x_1,\dots,x_n)=0)$$

Recall that we denote $(h_0(P),\dots,h_n(p))$ by the h-vector of P.

Corollary 10.2.2
Let M be a quasitoric manifold with characteristic pair (P,Λ). Then we have

$$\text{rank}(H^{2k}(M;\mathbb{Z}))=h_k(P)$$

10.3 Bundles over Quasitoric Manifolds

Proposition 10.3.1
Let M be a quasitoric manifold. Then $TM\oplus\mathbb{R}^k$ admits a complex structure for some $k\in\mathbb{N}$.

The chern class of $TM\oplus\mathbb{R}^k$ is given by

$$c(TM\oplus\mathbb{R}^k)=\prod_{i=1}^m(1+x_i)\in H^*(M)\cong\frac{\mathbb{Z}[x_1,\dots,x_m]}{I_{\text{SR}}+I_\Lambda}$$

There is a diffeo $Z_P/(G(P)\cong T^{m-n})\cong M(P,\Lambda)$ for any P. (from this perspective one can see that projective toric manifolds are quasi-toric manifolds). Also this map $Z_P\to M(P,\Lambda)$ is a principle $G(P)$-bundle.

If $M(P,\Lambda) = V_P$ is a projective toric variety, the classical construction $\mathbb{C}^m \setminus Z(\Sigma_P)$ exhibits Z_P algebraically: there is an equivariant homotopy equivalence $\mathbb{C}^m \setminus Z(\Sigma_P) \simeq Z_P$.

11 Kahler Manifolds

11.1 Equivalent Definitions

Definition 11.1.1 (Kahler Forms)
Let (M,h) be a complex Hermitian manifold. Let ω be the associated form. We say that ω is a Kahler form if $d\omega=0$. In this case, we say that M is a Kahler manifold.

Lemma 11.1.2
Let M be a Kahler manifold. Then the following are true.
- Let $g=\frac{1}{2}(h+\overline{h})$ be the associated Riemannian metric. Then (M,g) is a Riemannian manifold.
- Let $\omega=\frac{i}{2}(h-\overline{h})$ be the associated $(1,1)$-form. Then (M,ω) is a symplectic manifold.

Proposition 11.1.3
Let (M,ω) be a symplectic manifold. Then ω is a Kahler form if and only if M admits an integrable almost complex structure J. In this case, the following are true.
- The Riemannian metric is given by $g(u,v)=\omega(u,Jv)$.
- The Hermitian metric is given by $h=g-i\omega$.

Proposition 11.1.4
Let (M,g) be a Riemannian manifold of dimension $2n$. Then M admits a Kahler form if and only if $\text{Hol}(\nabla^g)\leq U(n)$.

Example 11.1.5
$\mathbb{CP}^n$ together with the Fubini-Study metric is a Kahler manifold.

Example 11.1.6
Every Riemann surfaces is a Kahler manifold.

11.2 Basic Structures

Theorem 11.2.1 (Banyaga)
Let M be a compact complex manifold. Let ω_1 and ω_2 be Kahler forms on M. Then (M,ω_1) and (M,ω_2) are symplectomorphic.

11.3 Main Theorems

Theorem 11.3.1 (Hodge Decomposition Theorem)
Let M be a compact Kahler manifold. Then the following are true.
- There is an isomorphism
$$H^k_{\text{dR}}(M;\mathbb{C})\cong\bigoplus_{p+q=k}H^{p,q}(M)$$
for all $k\in\mathbb{N}$.
- There is an isomorphism
$$H^{p,q}(M)\cong\overline{H^{q,p}(M)}$$
for all $p,q\in\mathbb{N}$.
- $\dim_{\mathbb{R}}(H^k_{\text{dR}}(M))$ is even for all $k\in\mathbb{N}$.

Example 11.3.2
Every compact Riemann surface is of the form Σ_g for some $g\geq 0$. We have seen above that Σ_g is

Kahler. We know that

$$H^k_{\text{dR}}(\Sigma_g;\mathbb{C}) = \begin{cases} \mathbb{C} & \text{if } k = 0, 2 \\ \mathbb{C}^{2g} & \text{if } k = 1 \\ 0 & \text{otherwise} \end{cases}$$

From this we conclude that $H^{0,0}(\Sigma_g) = \mathbb{C}$. Now since $\dim_{\mathbb{C}}(H^{p,q}(M)) = \dim_{\mathbb{C}}(\overline{H^{q,p}(\Sigma_g)}) = \dim_{\mathbb{C}}(H^{q,p}(\Sigma_g))$, we conclude that $H^{1,0}(\Sigma_g) \cong H^{0,1}(\Sigma_g) \cong \mathbb{C}^g$. Also, since Σ_g has complex dimension 1, it has no holomorphic 2-forms. Hence $H^{2,0}(\Sigma_g) = H^{0,2}(\Sigma_g) = 0$. This leaves us with $H^{1,1}(\Sigma_g) = H^2(\Sigma_g) = \mathbb{C}$. The hodge diamond of Σ_g is given by

$$\begin{array}{ccc} & 1 & \\ g & & g \\ & 1 & \end{array}$$

Definition 11.3.3 (Hodge Filtration)
Let M be a compact Kahler manifold. Define the Hodge filtration of M to be

$$F^i H^k_{\text{dR}}(M;\mathbb{C}) = \bigoplus_{p\geq i} H^{p,q}(M)$$

Definition 11.3.4 (The Lefschetz Operator)
Let M be a Kahler manifold. Let ω be its associated form. Define the Lefschetz operator to be the map $L : \Omega^k(M) \to \Omega^{k+2}(M)$ given by

$$\alpha \mapsto \alpha \wedge \omega$$

Theorem 11.3.5 (Hard Lefschetz Theorem)
Let M be a compact Kahler manifold. Then the following are true.

- The Lefschetz operator induces an isomorphism

$$L^k : H^{n-k}_{\text{dR}}(M) \xrightarrow{\cong} H^{n+k}_{\text{dR}}(M)$$

Definition 11.3.6 (Primitive Cohomology)
Let M be a compact Kahler manifold. Let $n = \dim(M)$. Define the primitive cohomology of M to be

$$H^i_{\text{prim}}(M) = \ker(L^{n-i+1} : H^i_{\text{dR}}(M) \to H^{2n-i+2}_{\text{dR}}(M))$$

Definition 11.3.7 (Lefschetz Decomposition)
Let M be a compact Kahler manifold. Define the Lefschetz decomposition of M to be

$$H^k_{\text{dR}}(M) = \bigoplus_{2r\leq k} L^r(H^{k-2r}_{\text{prim}}(M))$$

Proposition 11.3.8
Let M be a compact Kahler manifold of complex dimension $2n$. Suppose that $h = \sum_{p=0}^{2m}\sum_{q=0}^{2m}(-1)^p \dim_{\mathbb{C}}(H^{p,q}(M))$. Then the signature of the non-degenerate symmetric bilinear form $H^n(X;\mathbb{R}) \times H^n(X;\mathbb{R}) \to \mathbb{R}$ from Poincare duality is given by $(h, 2n-h)$.

11.4 Line Bundles on Kahler Manifolds

Theorem 11.4.1 (Lefschetz Theorem on $(1,1)$-Classes)
Let M be a compact Kahler manifold. Then we have

$$\text{im}(\text{Pic}(M) \to H^2(M,\mathbb{Z}) \to H^2(M,\mathbb{C})) = \text{im}(H^2(M,\mathbb{Z}) \to H^2(M,\mathbb{C})) \cap H^{1,1}(M)$$

12 Algebra

12.1 Algebra, Coalgebra and Bialgebra

Let R be a commutative ring. Recall that an R-algebra is a ring A together with a ring homomorphism $R \to A$. Equivalently, it is an R-module with multiplication that is associative and respects the R-module structure.

Proposition 12.1.1
Let R be a commutative ring. Let $A \in \mathbf{Mod}_R$ be an R-module. Let $m : A \otimes_R A \to A$ and $u : R \to A$ be R-module homomorphisms. Then (A, m, u) is an associative algebra with multiplication given by m and identity given $u(1)$ if and only if the following diagrams commute:

$$\begin{tikzcd}
(A \otimes_R A) \otimes_R A \arrow[r, "m\otimes 1"] \arrow[d, "\alpha"'] & A \otimes_R A \arrow[r, "m"] & A \\
A \otimes_R (A \otimes_R A) \arrow[r, "1\otimes m"'] & A \otimes_R A \arrow[ur, "m"'] &
\end{tikzcd}
\qquad
\begin{tikzcd}
R \otimes_R A \arrow[r, "u\otimes \mathrm{id}_A"] \arrow[dr, "\lambda"'] & A \otimes_R A \arrow[d, "m"'] & A \otimes R \arrow[l, "\mathrm{id}_A \otimes u"'] \arrow[dl, "\mu"] \\
 & A &
\end{tikzcd}$$

where α is the natural isomorphism from associativity of tensor products, and λ, μ is the natural isomorphism from the unitality of tensor products.

Equivalently, the above proposition is the same as saying (A, m, u) is a monoid object in the monoidal category $(\mathbf{Mod}_R, \otimes_R, R)$. Motivated by the categorical description of an R-algebra, we define coalgebras as the formal dual of the above object in the following way.

Definition 12.1.2 (Coalgebras)
Let R be a commutative ring. Let $C \in \mathbf{Mod}_R$ be an R-module. Let $\Delta : C \to C \otimes_R C$ and $\varepsilon : C \to R$ be R-module homomorphisms. We say that (C, Δ, ε) is a coalgebra if the following diagrams commute:

$$\begin{tikzcd}
C \arrow[r, "\Delta"] \arrow[dr, "\Delta"'] & C \otimes_R C \arrow[r, "\Delta\otimes 1"] & (C \otimes_R C) \otimes_R C \arrow[d, "\alpha"] \\
 & C \otimes_R C \arrow[r, "1\otimes\Delta"'] & C \otimes_R (C \otimes_R C)
\end{tikzcd}
\qquad
\begin{tikzcd}
 & C \arrow[dl, "\lambda"'] \arrow[d, "\Delta"] \arrow[dr, "\mu"] & \\
R \otimes_R C & C \otimes C \arrow[l, "\varepsilon\otimes \mathrm{id}_A"] \arrow[r, "\mathrm{id}_A \otimes \varepsilon"'] & C \otimes_R R
\end{tikzcd}$$

where α is the natural isomorphism from associativity of tensor products, and λ, μ is the natural isomorphism from the unitality of tensor products.

Definition 12.1.3 (Bialgebras)
Let R be a commutative ring. Let $B \in \mathbf{Mod}_R$ be an R-module. Let $m : B \otimes_R B \to B$, $u : R \to B$, $\Delta : B \to B \otimes_R B$ and $\varepsilon : B \to R$ be R-module homomorphisms. We say that B is an R-bialgebra if the following are true.

- (B, m, u) is an R-algebra.
- (B, Δ, ε) is an R-coalgebra.
- Multiplication and comultiplication are compatible:

$$\begin{tikzcd}
B \otimes_R B \arrow[r, "m"] \arrow[d, "\Delta\otimes\Delta"'] & B \arrow[r, "\Delta"] & B \otimes_R B \\
B \otimes_R B \otimes_R B \otimes_R B \arrow[rr, "\mathrm{id}\otimes\tau\otimes\mathrm{id}"'] & & B \otimes_R B \otimes_R B \otimes_R B \arrow[u, "m\otimes m"']
\end{tikzcd}$$

  where τ is the natural isomorphism from commutativity of tensor products.
- Multiplication and counit are compatible:

$$\begin{tikzcd}
B \otimes_R B \arrow[r, "m"] \arrow[d, "\varepsilon\otimes\varepsilon"'] & B \arrow[d, "\varepsilon"] \\
R \otimes_R R \arrow[r, "\lambda"'] & R
\end{tikzcd}$$

  where λ is the natural isomorphism from unitality of tensor products.

- Comultiplication and unit are compatible:

$$\begin{tikzcd} R \arrow[r, "u"] \arrow[d, "\lambda"'] & B \arrow[d, "\Delta"] \\ R \otimes_R R \arrow[r, "\Delta\otimes\Delta"'] & B \otimes_R B \end{tikzcd}$$

where λ is the natural isomorphism from unitality of tensor products.
- Unit and counit are compatible:

$$\begin{tikzcd} R \arrow[rr, "\mathrm{id}"] \arrow[dr, "u"'] & & R \\ & B \arrow[ur, "\varepsilon"'] & \end{tikzcd}$$

12.2 Hopf Algebra

Definition 12.2.1 (Hopf Algebras)
Let R be a commutative ring. Let $(H, m, u, \Delta, \varepsilon)$ be an R-bialgebra. Let $S : H \to H$ be an R-module homomorphism. We say that $(H, m, u, \Delta, \varepsilon, S)$ is a Hopf algebra if the following diagram commutes:

$$\begin{tikzcd} & H \otimes_R H \arrow[rr, "S\otimes \mathrm{id}_H"] & & H \otimes_R H \arrow[dr, "m"] & \\ H \arrow[ur, "\Delta"] \arrow[rr, "\varepsilon"] \arrow[dr, "\Delta"'] & & R \arrow[rr, "u"] & & H \\ & H \otimes_R H \arrow[rr, "\mathrm{id}_H\otimes S"'] & & H \otimes_R H \arrow[ur, "m"'] & \end{tikzcd}$$

Definition 12.2.2 (Commmutative and Cocommutative)
Let R be a commutative ring. Let H be a Hopf algebra.
- We say that H is commutative if the underlying algebra is commutative.
- We say that H is cocommutative if the underlying coalgebra is cocommutative.

Proposition 12.2.3
Let R be a commutative ring. Let H be a Hopf algebra. If H is a projective and finitely generated R-modue, then $\mathrm{Hom}_R(H, R)$ is a Hopf algebra. Moreover, the following are true.
- $\mathrm{Hom}_R(M, R)$ is commutative if and only if H is cocommutative.
- $\mathrm{Hom}_R(M, R)$ is cocommutative if and only if H is commutative.

12.3 Differential Graded Algebras

Definition 12.3.1 (Differential Graded Algebras)
Let R be a commutative ring. A differential graded R-algebra consists of the following data.
- A $\mathbb{Z}$-graded R-algebra A.
- An R-algebra homomorphism of degree -1 (called homological dgas) or of degree 1 (called cohomological dgas).

such that the following are true.
- Differential: $d \circ d = 0$.
- Graded Leibniz: $d(ab) = (da)b + (-1)^{\deg(a)}a(db)$ for a, b homogeneous elements of A.

The differential property means that any (homological) differential graded algebra decomposes into a chain complex whose differential is given by d.

Definition 12.3.2 (Homomorphisms of Differential Graded Algebras)
Let R be a commutative ring. Let A, B be differential graded algebras. A morphism $\varphi : A \to B$ of differential graded algebras is a graded R-algebra homomorphism.

Lemma 12.3.3
Let R be a commutative ring. Let (A, d) be a cohomological differential graded algebra. Then $H^*(A, d) = \bigoplus_{n\in\mathbb{Z}} H^n(A, d)$ is a graded ring with multiplication defined by

$$[u] \cdot [v] = [uv]$$

for $u \in H^n(A, d)$ and $v \in H^m(A, d)$.

Proof
Notice that by the graded Leibniz rule, we have $d(uv) = (du)v + (-1)^n ud(v) = 0$ since u and v are cocycles. Hence uv is also a cocycle. ■

Products of DGA, tensor products, hom set

Definition 12.3.4 (Commutative Differential Graded Algebras)
Let R be a commutative ring. Let A be a differential graded algebra. We say that A is commutative (cdga) if A is graded commutative.

Lemma 12.3.5
Let R be a commutative ring. Let M be an R-module. Let $\varphi : M \to R$ be an R-module homomrphism. Then $K_\bullet(\varphi)$ is a commutative differential graded algebra.

Proposition 12.3.6
Let k be a field. Let A, B be cohomological differential graded algebras. Then the following are true.
- There is a natural isomorphism $H^*(A) \otimes_k H^*(B) \cong H^*(M \otimes_k N)$.
- There is a natural isomorphism $H^*(\text{Hom}_k(A, B)) \cong \text{Hom}_k(H^*(A), H^*(B))$

12.4 Minimal DGAs

Definition 12.4.1 (Free Graded-Commutative Algebra)
Let R be a commutative ring. Let M be a graded R-module. Define the free graded-commutative algebra on M by

$$E(M) = \text{Sym}\left(\bigoplus_{i\in 2\mathbb{Z}} M_i\right) \oplus \Lambda\left(\bigoplus_{i\in 2\mathbb{Z}+1} M_i\right)$$

Definition 12.4.2
Let R be a ring. Let M be a graded R-module. A minimal DGA over M is a differential graded algebra $(E(M), d)$ such that $d(M) \subseteq E(M)_2$.

12.5 Massey Products

Definition 12.5.1 (Massey Triple Products)
Let R be a commutative ring. Let A be a differential graded algebra. Let $u \in A^p$, $v \in A^q$ and $w \in A^r$. Suppose that $[u] \cdot [v] = [v] \cdot [w] = 0$. Define the Massey triple product of the three elements to be

$$\langle [u], [v], [w] \rangle = \{[\overline{s} \cdot w + \overline{u} \cdot t] \in H^{p+q+r-1}(A) \mid s \in A^{p+q-1}, t \in A^{q+r-1} \text{ and } ds = \overline{u} \cdot v, dt = \overline{v} \cdot w\}$$

where $\overline{a} = (-1)^{1+\deg(a)} a$ for any homogeneous element $a \in A$.

Lemma 12.5.2
Let R be a commutative ring. Let A be a differential graded algebra. Let $u \in A^p$, $v \in A^q$ and $w \in A^r$. Suppose that $[u] \cdot [v] = [v] \cdot [w] = 0$. Then we have

$$\langle [u], [v], [w] \rangle = \frac{H^{p+q+r-1}(A)}{[u] \cdot H^{q+r-1}(A) + H^{p+q-1}(A) \cdot [w]}$$

Proposition 12.5.3
Let $p, q, r \in \mathbb{N}$. Let $Y = S^p \vee S^q \vee S^r$. Let $\alpha \in \pi_p(Y)$, $\beta \in \pi_q(Y)$ and $\gamma \in \pi_r(Y)$ be generators. Let $X = Y \cup_{[\alpha,[\beta,\gamma]]} D^{p+q+r-1}$. Let $u, v, w, z \in H^*(X)$ be generators of degree $p, q, r, p+q+r-1$ respectively. Then the following are true.

- $\langle u, v, w \rangle = (-1)^p z$.
- $\langle v, w, u \rangle = (-1)^{pq+pr+p} z$.
- $\langle w, u, v \rangle = 0$.

13 Koszul Algebras and Quadratic Algebras

14 Profinite Groups

14.1 Basic Properties

Definition 14.1.1 (Profinite Groups)
Let G be a topological group. We say that G is a profinite group if there exists an inverse system $X : \mathcal{I} \to {}_G\textbf{Top}$ consisting of finite groups with the discrete topology such that

$$G = \lim_{\mathcal{I}} X$$

In particular, this means that G is equipped with the weak topology.

Proposition 14.1.2
Let G be a topological group. Then G is profinite if and only if G is compact and totally disconnected.

Corollary 14.1.3
Let G be a profinite group. Then the following are true.
- G is Haudorff.
- G is locally compact.

Proposition 14.1.4
Let G be a compact topological group. Then G/G_0 is a profinite group.

Proposition 14.1.5
Let G be a profinite group. Let $H \leq G$ be a subgroup. Then the following are true.
- If H is closed, then H is profinite.
- If H is closed and normal, then G/H is profinite.
- H is open if and only if H is closed and $[G : H] < \infty$.

14.2 Profinite Completion

15 Equivariant Algebraic Topology

15.1 G-Homotopy

Definition 15.1.1 (G-Homotopy) Let G be a topological group and let X, Y be G-spaces. Let $f, g : X \to Y$ be G-equivariant maps. A G-homotopy from f to g is a homotopy $H : X \times I \to Y$ from f to g such that for each $t \in I$, the map

$$H(-, t) : X \to Y$$

is G-equivariant map.

15.2 Equivariant CW Complexes

15.3

Definition 15.3.1 (The Homotopy Quotient (Borel Construction))
Let G be a topological group. Let X be a G-space. Define the homotopy quotient of X to be the orbit space

$$EG \times_G X = \frac{EG \times X}{G}$$

where the action of G on $EG \times X$ is given by $g \cdot (e, x) = (g \cdot e, g^{-1} \cdot x)$.

Proposition 15.3.2
Let G be a topological group. Let X be a G-space. Then

$$X \hookrightarrow EG \times_G X \to BG$$

is a fibration.

Lemma 15.3.3
Let G be a topological group. Let X be a G-space. If G acts freely on X, then there is a homotopy equivalence

$$EG \times_G X \simeq \frac{X}{G}$$

induced by the projection map $EG \times_G X \to X$.

16

Definition 16.0.1 (Gauge Group)
Let G be a topological group. Let $p : E \to X$ be a principal G-bundle. Define the gauge group of p to be the set

$$G(p) = \{f : E \to E \mid f \text{ is an isomorphism of } p\}$$

together with composition of functions.

Proposition 16.0.2
Let G be a topological group. Let $p : E \to X$ be a principal G-bundle. Then there is a group isomorphism

$$G(p) \cong \Gamma(\widetilde{p})$$

where $\widetilde{p}$ is the associated principal bundle where G acts on the fibers by the adjoint representation.

Note: The associated principal bundle is the space

$$\frac{E \times G}{\sim}$$

where $(p, g) \sim (ph^{-1}, hgh^{-1})$ for $p \in E$ and $g, h \in G$.

17

There is a one to one bijection between representations of $\pi_1(X)$ and locally constant sheaves (called local systems).

Definition 17.0.1
Let X be a space. Let $x_0 \in X$. Define the functor $\text{Fib}_{x_0} : \text{Cov}(X) \to {}_{\pi_1(X,x_0)}\textbf{Sets}$ as follows.

- For $p : \widetilde{X} \to X$ a covering space, define $\text{Fib}_{x_0}(p) = p^{-1}(x_0)$.
- For $f : \widetilde{X}_1 \to \widetilde{X}_2$ a map of covering spaces, define $\text{Fib}_{x_0}(f)$ to be $f|_{p^{-1}(x_0)}$.

Theorem 17.0.2
Let X be a connected and locally simply connected space. Let $x_0 \in X$. Then

$$\text{Fib}_{x_0} : \text{Cov}(X) \to {}_{\pi_1(X,x_0)}\textbf{Sets}$$

is fully faithful and essentially surjective.

Theorem 17.0.3 (Classification of Local Systems)
Let X be a space (locally compact, Hausdorff, second countable, locally path connected, semi locally path connected, path connected). Then there is an equivalent of categories

$$\text{Func}(\textbf{Open}^{\text{op}}, \text{Vect}_{\mathbb{C}}) \simeq \text{Rep}(\pi_1(X))$$

Ref: Galois Groups and Fundamental Groups

Theorem 17.0.4 (Classification Theorem for Flat Connections)
Let M be a connected Riemannian manifold. Let G be a Lie group. Then there is a one to one correspondence

$$\frac{\{(P, A) \mid P \to M \text{ is a principal } G\text{-bundle and } A \text{ is a flat connection}\}}{\sim} \overset{1:1}{\longleftrightarrow} \text{Rep}(\pi_1(X), G)$$

where $(P_1, A_1) \sim (P_2, A_2)$ if there exists bundle isomorphism $\varphi : P_1 \to P_2$ such that $\varphi^* A_2 = A_1$.

Ref: Differential Geometry; Taubes Bijection described in: Geometry of Characteristic classes; Morita

There is a natural bijection between principal bundles and vector bundles by taking the associated bundle. If $G = GL(V)$ for some finite dimensional vector space V over the field $k = \mathbb{R}$ or $\mathbb{C}$, the sheaf of sections of the associated bundle is a locally constant sheaf.

Also: Upgrade to diffeomorphism.

18 Infinite Galois Theory

Proposition 18.0.1
Let F be a field. Let K/F be a Galois extension. Then the following are true.

- The inverse limit of the system $\text{Gal}(L/F)$ for L/F a finite Galois extension is $\text{Gal}(K/F)$.
- $\text{Gal}(K/F)$ is a profinite group.
- The projection maps $\text{Gal}(K/F) \to \text{Gal}(L/F)$ for L/F a finite Galois extension is a surjective homomorphism.
- Let $F < L < K$ be a field extension. Then K/L is a Galois extension and $\text{Gal}(K/L)$ is a closed subgroup of $\text{Gal}(K/F)$.

Theorem 18.0.2 (Krull)

Let F be a field. Let K/F be a field extension. Then there is an inclusion reversing bijection

$$\begin{Bmatrix} \text{Closed subgroups of} \\ \text{Aut}_F(K) \end{Bmatrix} \overset{1:1}{\longleftrightarrow} \begin{Bmatrix} \text{Intermediary subfields} \\ F < L < K \end{Bmatrix}$$

given as follows.

- For a subgroup $H \leq \text{Aut}_F(K)$, $H \mapsto \text{Fix}_F(H)$.
- For an intermediary subfield M, $M \mapsto \text{Aut}_L(K)$.

Proposition 18.0.3

Let F be a field. Let K/F be a Galois extension. Then the above bijection restricts to a bijection

$$\begin{Bmatrix} \text{Normal Subgroups} \\ \text{of Aut}_K(L) \end{Bmatrix} \overset{1:1}{\longleftrightarrow} \begin{Bmatrix} \text{Normal Extensions} \\ K < M \end{Bmatrix}$$

19 Local Cohomology

19.1

Definition 19.1.1 (The I-Torsion of a Module)
Let R be a Noetherian commutative ring. Let M be an R-module. Let I be an ideal of R. Define the I-torsion of M to be

$$\Gamma_I(M) = \{m \in M \mid I^t m = 0 \text{ for some } t \in \mathbb{N}\}$$

Definition 19.1.2 (I-Torsion Modules)
Let R be a Noetherian commutative ring. Let M be an R-module. Let I be an ideal of R. We say that M is I-torsion if $M = \Gamma_I(M)$.

Definition 19.1.3 (The Torsion Functor)
Let R be a Noetherian commutative ring. Let I be an ideal of R. Define the I-torsion functor

$$\Gamma_I : {}_R\textbf{Mod} \to {}_R\textbf{Mod}$$

by the following.
- For each R-module M, the functor sends M to the I-torsion $\Gamma_I(M)$.
- For each R-module homomorphism $f : M \to N$, the functor sends f to $\Gamma_I(f) = f|_{\Gamma_I(M)}$.

Lemma 19.1.4
Let R be a Noetherian commutative ring. Let I be an ideal of R. Then Γ_I is a left exact functor.

Definition 19.1.5 (Local Cohomology)
Let R be a Noetherian commutative ring. Let M be an R-module. Let I be an ideal of R. Define the nth local cohomology of M with support in I to be

$$H_I^n(M) = (R^n\Gamma_I)(M)$$

Proposition 19.1.6
Let R be a Noetherian commutative ring. Let M be an R-module. Let I, J be ideals of R. If $\sqrt{I} = \sqrt{J}$, then the equality $\Gamma_I(M) = \Gamma_J(M)$ induces an equality

$$H_I^n(M) = H_J^n(M)$$

for all $n \in \mathbb{N}$.

Thus we only need to consider the local cohomology with support in radical ideals.

Lemma 19.1.7
Let R be a Noetherian commutative ring. Let M be an R-module. Let I be an ideal of R. Then $H_I^n(M)$ is I-torsion for all $n \in \mathbb{N}$.

Lemma 19.1.8
Let R be a Noetherian commutative ring. Let M_i be R-modules for each $i \in J$. Let I be an ideal of R. Then there is a natural isomorphism

$$H_I^n\left(\bigoplus_{i \in J} M_i\right) \cong \bigoplus_{i \in J} H_J^n(M_i)$$

Proposition 19.1.9
Let R be a Noetherian commutative ring. Let M be an R-module. Let I be an ideal of R.Then there are natural isomorphisms

$$\varinjlim_{i\in\mathbb{N}} \text{Ext}_R^n(R/I^i, M) \cong H_I^n(M)$$

for all $n \in \mathbb{N}$.

20 Morita Theory

20.1

Definition 20.1.1 (Morita Equivalence)
Let R, S be rings. We say that R and S are Morita equivalent if there exists an equivalence of categories
$$_R\mathbf{Mod} \simeq {}_S\mathbf{Mod}$$

Lemma 20.1.2
Let R, S be rings. Then R and S are Morita equivalent if and only if there exists an equivalence of categories
$$\mathbf{Mod}_R \simeq \mathbf{Mod}_S$$

Lemma 20.1.3
Let R, S be rings. Suppose that R and S are Morita equivalent. Let $F : {}_R\mathbf{Mod} \to {}_S\mathbf{Mod}$ be an equivalence. Then F is additive.

20.2 Progenerators

Definition 20.2.1 (Progenerators)
Let R be a ring. Let P be an R-module. We say that P is a progenerator of R if the following are true.
- P is finitely generated.
- P is projective.

Proposition 20.2.2
Let R, S be rings. Let $F : {}_R\mathbf{Mod} \to {}_S\mathbf{Mod}$ and $G : {}_S\mathbf{Mod} \to {}_R\mathbf{Mod}$ be additive functors. Then F and G are inverses if and only if there exists an (S, R)-bimodule P such that the following are true.
- ${}_SP$ is a progenerator.
- P_R is a progenerator.
- There are natural isomorphisms $F \cong P \otimes_R -$ and $G \cong \mathrm{Hom}({}_SP, -)$.

Theorem 20.2.3 (Morita)
Let R, S be rings. Then the following are equivalent.
- R and S are Morita equivalent.
- There exists a progenerator R-module P_R such that $S \cong \mathrm{End}_R(P_R)$.
- There exist an (R, S)-bimodule M and a (S, R)-bimodule N such that $M \otimes_S N \cong R$ as (R, R)-bimodules and $N \otimes_R M \cong S$ as (S, S)-bimodules.

20.3 Properties of Morita Equivalence

Proposition 20.3.1
Let R, S be rings. If R and S are Morita equivalent, then the following are true.
- There is a ring isomorphism $Z(R) \cong Z(S)$.
- $R/J(R)$ is Morita equivalent to $S/J(S)$.

Proposition 20.3.2
Let R, S be rings. Let M be an R-module. Suppose that R and S are Morita equivalent via the functor $F : {}_R\mathbf{Mod} \to {}_S\mathbf{Mod}$. Then the following are true.
- M is injective if and only if $F(M)$ is injective.
- M is projective if and only if $F(M)$ is projective.
- M is simple if and only $F(M)$ is simple.

- M is semisimple if and only if $F(M)$ is semisimple.
- M is finitely generated if and only if $F(M)$ is finitely generated.
- M is Noetherian if and only if $F(M)$ is Noetherian.
- M is Artinian if and only if $F(M)$ is Artinian.

Proposition 20.3.3
Let R, S be rings. Suppose that R and S are Morita equivalent. Then the following are equivalent.
- R is simple if and only if S is simple.
- R is semisimple if and only if S is semisimple.
- R is Noetherian if and only if S is Noetherian.
- R is Artinian if and only if S is Artinian.

Proposition 20.3.4
Let R, S be rings. Then R and S are Morita equivalent if and only if ${}_R\mathbf{FGMod} \simeq {}_S\mathbf{FGMod}$.

21 Complex Analytic Space

21.1 Complex Analytic Varieties

Definition 21.1.1 (Complex Analytic Varieties)
Let $Z \subseteq \mathbb{C}^n$. We say that Z is a complex analytic variety if there exists an open set $Z \subseteq U \subseteq \mathbb{C}^n$ and $f_1, \dots, f_k \in \mathcal{O}_{\mathbb{C}^n}(U)$ such that
$$Z = \mathbb{V}(f_1, \dots, f_k)$$

We use a subset of C^n in the definition because in $\mathbb{C}^n$, locally defined functions cannot in general be defined globally (compare it to the real case, where they have partitions of unity and bump functions).

Indeed, for smooth functions on $\mathbb{R}^n$, a function on an open set U can be modified (via bump functions) to produce a global function that agrees with it on a smaller open subset. You can always find a global function that captures the local behavior near any point.

For holomorphic functions on $\mathbb{C}^n$: no such modification is possible. A holomorphic function cannot be cut off or modified while remaining holomorphic, so local holomorphic data is genuinely trapped in its domain.

Definition 21.1.2 (The Structure Sheaf) Let$U \subseteq \mathbb{C}^n$ be an open subset. Let $f_1, \dots, f_k \in \mathcal{A}_{\mathbb{C}^n}(U)$ be analytic functions on U. Let $Z = \mathbb{V}(f_1, \dots, f_k)$ be the vanishing locus of $f_1, \dots, f_k$. Define the structure sheaf
$$\mathcal{O}_Z : \textbf{Open}(Z) \to \textbf{CRing}$$
as follows.

- For each open set $U \subseteq Z$, define
$$\mathcal{O}_Z(U) = \frac{\mathcal{O}_{\mathbb{C}^n}(U)}{(f_1, \dots, f_k)}$$
- For each inclusion $U \subseteq V \subseteq Z$ of open sets, define
$$\mathcal{O}_Z(V) \to \mathcal{O}_Z(U)$$
to be the induced map of the restriction $\mathcal{O}_{\mathbb{C}^n}(V) \to \mathcal{O}_{\mathbb{C}^n}(U)$.

21.2 Complex Analytic Subvarieties of Manifolds

Definition 21.2.1 (Complex Analytic Subvarieties)
Let M be a complex manifold. Let $Z \subseteq M$ be a subset. We say that Z is a complex analytic subvariety of M if for each $p \in Z$, there exists a chart (U, ϕ) of M such that $\phi(Z \cap U)$ is a complex analytic variety.

In particular, complex analytic subvarieties of $\mathbb{C}^n$ are just complex analytic varieties.

Theorem 21.2.2 (Cartan's (Coherence) Theorem) Let M be a complex manifold and let A be an analytic subset of M. Then $\mathcal{I}_A$ is a coherent sheaf of $\mathcal{O}_M$-modules.

21.3 Complex Analytic Spaces

Definition 21.3.1 (Complex Analytic Spaces) Let $(X, \mathcal{O}_X)$ be a locally ringed space. We say that X is a complex analytic space if for all $p \in X$, there exists an open neighbourhood $U \subseteq X$ such that $(U, \mathcal{O}_X|_U)$ is a complex analytic space.

Lemma 21.3.2
Let $(X, \mathcal{O}_X)$ be a complex analytic space. Then X is a complex manifold if and only if for each $p \in X$, there exists an open neighbourhood $U \subseteq X$ such that there is an isomorphism

$$(U, \mathcal{O}_X|_U) \cong (V, \mathcal{O}_{\mathbb{C}^n}|_V)$$

for some open subset $V \subseteq \mathbb{C}^n$.

Theorem 21.3.3 Let X be a complex space. Then X_{reg} is dense and open in X. Moreover, X_{reg} consists of a disjoin union of complex manifolds.

Theorem 21.3.4 Let X be an irreducible complex space. Then every non-constant holomorphic function $f : X \to \mathbb{C}$ is an open map.

Corollary 21.3.5 Let X be an irreducible compact complex space. Then every holomorphic function $f : X \to \mathbb{C}$ on X is constant.

Definition 21.3.6 (Complex Analytic Subvarieties)
Let $(X, \mathcal{O}_X)$ be a complex analytic space. Let $Z \subseteq X$ be subset. We say that Z is a complex analytic subvariety if for all $p \in X$, there exists an open neighbourhood $U \subseteq X$ and $f_1, \dots, f_k \in \mathcal{O}_X(U)$ such that

$$Z \cap U = \mathbb{V}_U(f_1, \dots, f_k)$$

This definition generalizes complex analytic subvarieties of complex manifolds.

21.4 Connections to Algebraic Geometry

Recall that the category of schemes and the category of analytic spaces are both full subcategories of locally ringed spaces.

Definition 21.4.1 (The Analytification of a Scheme)
Let X be a scheme locally of finite type over $\mathbb{C}$. Define the analytification of X to be the unique analytic space X^{an} together with a morphism of locally ringed space $\alpha : X^{\text{an}} \to X$ satisfying the following universal property: For any analytic space Y and any morphism of locally ringed spaces $f : Y \to X$, there exists a unique morphism of analytic spaces $g : Y \to X^{\text{an}}$ such that the following diagram commutes:

$$\begin{tikzcd} & Y \arrow[dl, dashed, "\exists ! g"'] \arrow[d, "f"] \\ X^{\text{an}} \arrow[r, "\alpha"'] & X \end{tikzcd}$$

Recall that the Yoneda embedding $h : \textbf{LRS} \to \textbf{Set}^{\textbf{LRS}}$ is the functor sending a locally ringed space X to the functor $h_X : \textbf{LRS} \to \textbf{Set}$ defined by $h_X(Y) = \text{Hom}_{\textbf{LRS}}(Y, X)$. Then the universal property of the analytification means that the functor h_X restricted to analytic spaces $\textbf{An}$ is representable.

One thinks of the morphism $X^{\text{an}} \to X$ as doing two things. It is an enlargement of the standard topology of X^{an} to the Zariski topology of X, and it introduces generic points outside of $X(\mathbb{C})$ (Indeed we will see that X^{an} has the same underlying set as $X(\mathbb{C})$).

Theorem 21.4.2
Let X be a scheme locally of finite type over $\mathbb{C}$. Then the analytification of X exists and is unique up to unique isomorphism.

Definition 21.4.3 (The Analytification Functor)
Define the analytification functor
$$(-)^{\text{an}} : \mathbf{Sch}_{\mathbb{C}} \to \mathbf{An}$$
as follows.

- For each scheme X locally of finite type over $\mathbb{C}$, we have X^{an} is the analytification of X.
- For a morphism of schemes $f : X \to Y$ over $\mathbb{C}$, the universal propertyof Y^{an} induces a morphism $X \to Y^{\text{an}}$. Precomposing with the canonical morphism $X^{\text{an}} \to Y^{\text{an}}$ gives the desired map $X^{\text{an}} \to Y^{\text{an}}$.

Proposition 21.4.4
Let X be a scheme locally of finite type over $\mathbb{C}$. Then the underlying set of X^{an} is given by $X(\mathbb{C})$.

Proposition 21.4.5
Let X be a scheme locally of finite type over $\mathbb{C}$. Then the following are true.

- X is separated if and only if X^{an} is Hausdorff.
- X is connected if and only if X^{an} is connected.
- X is smooth if and only if X^{an} is a complex manifold.
- X is proper if and only if X^{an} is compact.

Proposition 21.4.6
Let X be a variety over $\mathbb{C}$. Then the following are true.

- If X is irreducible, then X^{an} is connected.
- If X is smooth and affine, then X^{an} is a Stein manifold.
- If X is smooth and projective, then X^{an} is Kahler.

Theorem 21.4.7 (GAGA)
Let X be a proper scheme of finite type over $\mathbb{C}$. Then the following are true.

- Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules. Then $\mathcal{F}^{\text{an}}$ is a sheaf of $\mathcal{O}_{X^{\text{an}}}$-module.
- Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules. There is a natural isomorphism
$$H^i(X, \mathcal{F}) \cong H^i(X^{\text{an}}, \mathcal{F}^{\text{an}})$$
- The analytification functor induces an equivalence of categories
$$\mathbf{Coh}_{\mathcal{O}_X} \cong \mathbf{Coh}_{\mathcal{O}_{X^{\text{an}}}}$$
- Let $Y \subseteq X^{\text{an}}$ be a closed complex analytic subspace. Then there exists a closed subvariety Z of X such that $Z^{\text{an}} = Y$.

Corollary 21.4.8 (Chow's Theorem)
Let X be a closed complex analytic subspace of $\mathbb{CP}^n$. Then X is a projective variety over $\mathbb{C}$.

The important point is that: the theory of smooth projective complex varieties are subsumed by the theory of Kahler manifolds. Divisors and line bundles correspond bijectively thanks to Chow's theorem.

Theorem 21.4.9 (Grothendieck)
Let X be a smooth variety over $\mathbb{C}$. Then there is a natural isomorphism
$$H^n_{\text{dR}}(X) \cong H^n_{\text{Holo}}(X^{\text{an}}) \cong H^n_{\text{dR}}(X^{\text{an}})$$
induced by the canonical map $\alpha : X^{\text{an}} \to X$.

Corollary 21.4.10
Let X be a smooth variety over $\mathbb{C}$. Then the above isomorphism and de Rham's theorem induces

a natural isomrophism

$$H^n_{\mathrm{dR}}(X) \cong H^n_{\mathrm{sing}}(X^{\mathrm{an}}; \mathbb{C})$$

Theorem 21.4.11 (The Period Isomorphism)
Let k be a subfield of $\mathbb{C}$. Let X be a smooth variety over k. Then there is a natural isomorphism

$$H^*_{\mathrm{dR}}(X) \otimes_k \mathbb{C} \cong H^*_{\mathrm{sing}}(X^{\mathrm{an}}; \mathbb{Q}) \otimes_{\mathbb{Q}} \mathbb{C}$$

22 Divisors and Line Bundles

22.1 Divisors

Definition 22.1.1 (Divisors on a Complex Manifold)
Let X be a complex manifold. A divisor on X is a formal linear combination

$$D = \sum_{\substack{V \text{ is an analytic variety of } X \\ \text{of dimension } \dim(X)-1}} n_V \cdot V$$

where $n_V \in \mathbb{Z}$, and such that for each $p \in X$, there exists an open neighbourhood U of p such that U meets finitely many V in the summand.

The locally finite condition here is required because in Algebraic Geometry, Noetherianess is assumed, otherwise they are the same.

Definition 22.1.2 (Group of Divisors)
Let X be a complex manifold. Define the group of divisors of X to be the set

$$\text{Div}(X) = \{D \mid D \text{ is a divisor on } X\}$$

together with addition given as a subgroup of $\prod_V \mathbb{Z} \cdot V$.

Definition 22.1.3 (Effective Divisors)
Let X be a complex manifold. Let $D = \sum_V n_V \cdot V \in \text{Div}(X)$ be a divisor. We say that D is effective if $n_V \geq 0$ for all V.

Definition 22.1.4 (Divisors of Meromorphic Functions)
Let X be a complex manifold. Let $f \in \mathcal{M}_X^*$ be a meromorphic function. Define the divisor of f to be

$$\text{div}(f) = \sum_V \nu_V(f) \cdot V$$

where ν_V is the valuation of $\mathcal{O}_{X,V}$.

Definition 22.1.5 (Cartier Divisors on a Complex Manifold)
Let X be a complex manifold. Define the group of divisors of X to be

$$\text{CDiv}(X) = H^0(X, \mathcal{M}_X^*/\mathcal{O}_X^*) = \Gamma(X, \mathcal{M}_X^*/\mathcal{O}_X^*)$$

Proposition 22.1.6
Let X be a complex manifold. Then there is a natural isomorphism

$$\text{Div}(X) \cong \text{CDiv}(X)$$

22.2 Holomorphic Line Bundles

Proposition 22.2.1
Let X be a complex manifold. Then the following diagram is commutative:

$$\begin{tikzcd}
\text{CDiv}(X) \arrow[r] \arrow[d, "\cong"'] & \text{Pic}(X) \arrow[d, "\cong"] \\
H^0(X, Q_X^*/\mathcal{O}_X^*) \arrow[r] & H^1(X, \mathcal{O}_X^*)
\end{tikzcd}$$

where the bottom horizontal map is induced by the long exact sequence of the short exact sequence $0 \to \mathcal{O}_X^* \to \mathcal{M}_X^* \to \frac{\mathcal{M}_X^*}{\mathcal{O}_X^*} \to 0$.

By Serre's correspondence, holomorphic line bundles correspond to locally free sheaves of $\mathcal{O}_M$-modules of rank 1, and recall isomorphism classes of these objects form a group structure with the tensor product, and it is group isomorphic to $H^1(M, \mathcal{O}_M^*)$.

Definition 22.2.2 (The Exponential Sequence)
Let M be a complex manifold. Define the exponential sequence of M to be the short exact sequence of sheaves of $\mathcal{O}_M$-modules given by

$$0 \longrightarrow \underline{\mathbb{Z}} \hookrightarrow \mathcal{O}_M \xrightarrow{f \mapsto e^{2\pi i f}} \mathcal{O}_M^* \longrightarrow 0$$

Proposition 22.2.3
Let M be a complex manifold. Let L be a holomorphic line bundle on M. Then the first chern class of L is given by the image of L in the homomorphism

$$\text{Pic}(M) \cong H^1(M, \mathcal{O}_M^*) \to H^2(M, \underline{\mathbb{Z}}) \cong H^2(M, \mathbb{Z})$$

induced by the long exact sequence of the exponential sequence.

22.3 Kodaira Dimension

Definition 22.3.1
Let M be a complex manifold. Let L be a holomorphic line bundle on M. Define ??? to be

$$R(M, L) = \bigoplus_{m=0}^{\infty} H^0(M, L^{\otimes m})$$

Definition 22.3.2 (The Canonical Ring)
Let M be a complex manifold. Define the canonical ring of M to be

$$R(M) = R(M, \mathcal{M}_M) = \bigoplus_{m=0}^{\infty} H^0(M, \mathcal{M}_M^{\otimes m})$$

Lemma 22.3.3
Let M be a complex manifold. If M is connected, then $R(M)$ is an integral domain.

Definition 22.3.4 (Kodaira Dimension)
Let M be a connected complex manifold. Define the Kodaira dimension of M to be

$$\text{kod}(M) = \begin{cases} -\infty & \text{if } R(X) = \mathbb{C} \\ \text{trdeg}_{\mathbb{C}}(Q(R(X))) - 1 & \text{otherwise} \end{cases}$$

23 Compact Kahler Manifolds

23.1 Kodaira Embedding Theorem

Proposition 23.1.1
Let X be a smooth projective variety over $\mathbb{C}$. Let $\mathcal{L}$ be a line bundle over X. Then $\mathcal{L}$ is ample if and only if $\mathcal{L}^{\text{an}}$ is positive.

Theorem 23.1.2 (Kodaira Embedding)
Let X be a compact complex manifold. Then the following are equivalent.

- X admits a holomorphic embedding into $\mathbb{CP}^n$.
- X admits an ample line bundle.
- X admits a positive line bundle.
- X is admits a Kahler metric ω such that $[\omega] \in H^{1,1}(X, \mathbb{Z})$.
- There exists a projective variety Z over $\mathbb{C}$ such that $Z^{\text{an}} = X$.

Corollary 23.1.3
There is an equivalence of categories between smooth projective varieties over $\mathbb{C}$ and compact Kahler manifolds given by the analytification functor.

Theorem 23.1.4 (Kodaira-Nakano Vanishing Theorem)
Let X be a compact Kahler manifold. Let $p : L \to X$ be a positive line bundle. Then we have

$$H^q(X, \Omega^p(L)) = 0$$

for $p + q > \dim(X)$.

Theorem 23.1.5 (Lefschetz Hyperplane Theorem)
Let X be a compact Kahler manifold. Let $V \subseteq X$ be a smooth variety of dimension $\dim(X) - 1$. Then the restriction map

$$H^k(X, \mathbb{Z}) \to H^k(V, \mathbb{Z})$$

induces isomorphisms for $0 \leq k < \dim(X) - 1$ and is injective for $k = \dim(X) - 1$.

Theorem 23.1.6 (The Lefschetz $(1, 1)$ Theorem)
Let X be a compact Kahler manifold. Let $\alpha \in H^2(X, \mathbb{Z})$. Then α is the first Chern class of a line bundle if and only if $\alpha \in H^{1,1}(X)$.

23.2 Projective Curves

Proposition 23.2.1
Let X be a smooth projective curve over $\mathbb{C}$. Then X^{an} is homeomorphic to $\Sigma_{p_g(X)}$ the orientable surface of genus $p_g(X)$.

Corollary 23.2.2
There is an equivalence of categories between smooth projective curves over $\mathbb{C}$ and compact Riemann surfaces.

24 Analytic Manifolds

24.1 Real Analytic Manifolds

Definition 24.1.1 (Real Analytic Manifolds) A real analytic manifold is a topological manifold with analytic transition maps.

In the real case, every analytic manifold is a smooth manifold because every analytic function is necessarily infinitely differentiable. While not every analytic function is smooth, we can still equip a smooth structure on analytic manifolds.

Theorem 24.1.2 ((Grauert-Whitney)) Every smooth manifolds admits a compatible real analytic structure.

Definition 24.1.3 (Analytic Functions) Let M be a real analytic manifold. Let $U \subseteq M$ be a subset. Let $f : U \to \mathbb{R}$ be a function. We say that f is analytic if for all $x \in U$, there exists a chart $(V, \varphi = x^1, \dots, x^n))$ of M such that

$$f \circ \varphi^{-1} : \mathbb{R}^n \to \mathbb{R}$$

is analytic in the usual sense.

Definition 24.1.4 (Sheaf of Analytic Functions) Let M be an analytic manifold. Define the sheaf of analytic functions

$$\mathcal{A}_M : \textbf{Open} \to \textbf{CRing}$$

as follows.

- For each open set $U \subseteq M$, $\mathcal{A}_M(U)$ consists of the ring of analytic functions on U
- For each inclusion $V \subseteq U$, there is a unique morphism

  $$\text{res}_V^U : \mathcal{A}_M(U) \to \mathcal{A}_M(V)$$

  given by $f \mapsto f|_V$

25 Analytic Varieties

25.1 Real Analytic Varieties

25.2 Regular and Singular Points

Definition 25.2.1 (Regular and Singular Points) Let M be a complex manifold and let A be an analytic subset of M. We say that $x \in A$ is a regular point if there exists some open neighbourhood U of x such that $A \cap U$ is a complex submanifold of M. Otherwise, x is said to be singular. Denote

$$A_{\text{reg}} = \{x \in A \mid x \text{ is a regular point }\} \quad \text{and} \quad A_{\text{sing}} = \{x \in A \mid x \text{ is a singular point }\}$$

Theorem 25.2.2 Let M be a complex manifold and let A be an analytic subset of M. Then A_{sing} is an analytic subset of A.

26 Chapter: The Geometry of Fiber Bundles

27 Fundamental Constructions

27.1 Properties of the Space of Vector Fields

Definition 27.1.1 (The Lie Bracket)
Let M be a smooth manifold. Let $X, Y \in \mathfrak{X}(M)$ be smooth vector fields. Define the Lie bracket of X and Y to be the vector field $[X, Y] \in \mathfrak{X}(M)$ given by

$$[X, Y]_p(f) = X_p(Y_p(f)) - Y_p(X_p(f))$$

for all $f \in C^\infty(M)$ for each $p \in M$.

Lemma 27.1.2
Let M be a smooth manifold. Then $(\mathfrak{X}(M), [-, -])$ is a Lie algebra.

Proposition 27.1.3
Let M be a smooth manifold. Let G be a Lie group acting smoothly on M. Then the map $\text{Lie}(G) \to \mathfrak{X}(M)$ given by

$$\xi \mapsto \xi_M$$

is a Lie algebra homomorphism.

27.2 The Normal Bundle

Definition 27.2.1 (The Normal Bundle)
Let M, N be smooth manifolds. Let $\iota : N \hookrightarrow M$ be a smooth embedding. Define the normal bundle of N in M to be the quotient bundle

$$N_{M/N} = \frac{TM}{TN}$$

Example 27.2.2
Let $\iota : S^n \to \mathbb{R}^{n+1}$ be the standard inclusion of the unit sphere. Then the following are true.

- $N_{\mathbb{R}^{n+1}/S^n} \cong \varepsilon$ is a trivial bundle of rank 1.
- $w_i(TS^n) = 1$ for all i.

Proof
Notice that $N_{\mathbb{R}^{n+1}/S^n}$ is a bundle of rank equal to 1. It is the trivial bundle if and only if it admits a nowhere vanishing section. Define a section $s : S^n \to N_{\mathbb{R}^{n+1}/S^n}$ by $s(x) = (x, x)$ where $x \in N_{\mathbb{R}^{n+1}/S^n}$ is a vector in the direction x. Then clearly it is non-vanishing.

Now we have $TS^n \oplus N_{\mathbb{R}^{n+1}/S^n} = T\mathbb{R}^{n+1}$. Applying the total Stiefel-Whitney class gives $w(TS^n) \cdot 1 = w(TS^n)w(N_{\mathbb{R}^{n+1}/S^n}) = w(T\mathbb{R}^{n+1}) = 1$. Hence $w(TS^n) = 1$. Since $w_0(TS^n) = 1$, the grading in the cohomology ring implies that $w_i(TS^n) = 0$ for all i. ∎

28 Smooth Fiber Bundles

28.1 Basic Structures

Definition 28.1.1 (Smooth Fiber Bundles)
Let E, M, F be smooth manifolds. Let $p : E \to M$ be a fiber bundle with fiber F and local trivialization charts (U_k, ϕ_k) for $k \in I$. We say that p is a smooth fiber bundle if the following are true.

- p is smooth.
- ϕ_k is a diffeomorphism for all $k \in I$.

Lemma 28.1.2
Let $p : E \to M$ be a smooth fiber bundle with fiber F. Let $p, q \in M$. If M is connected, then E_p is diffeomorphic to E_q.

Definition 28.1.3 (Smooth Sections)

28.2 The Jet Bundle

Definition 28.2.1 (Differentiating Smooth Sections)
Let $\pi : E \to M$ be a smooth fiber bundle with fiber F. Let $p \in M$. Let $(U, \theta = (x^1, \dots, x^{\dim(M)}))$ be a local chart of M at p. Let $s : U \to E$ be a local section. Let (V, ϕ) be a local trivialization chart of π containing $s(x)$. Let (W, ψ) be a local chart of F at $\text{proj}_F(\phi(s(p)))$. Define the derivative of s at p with respect to x^k by to be

$$\left.\frac{\partial s}{\partial x^k}\right|_p = \left.\frac{\partial(\psi \circ \text{proj}_F \circ \phi \circ s \circ \theta^{-1})}{\partial x_k}\right|_{\theta(p)}$$

Prove: independent of choice of chart

Definition 28.2.2 (Jet Equivalence)
Let $\pi : E \to M$ be a smooth fiber bundle with fiber F. Let $p \in M$. Let $U_1, U_2 \subseteq M$ be subsets such that $p \in U_1, U_2$. Let $s_1 : U_1 \to E$ and $s_2 : U_2 \to E$ be smooth sections. We say that s_1 and s_2 are k-jet equivalent at p if the following are true.

- $s_1(p) = s_2(p)$.
- There exist a chart (U, θ) of M at p, a trivialization (A, ϕ) of π over an open set containing p, and a chart (W, ψ) of F at $\text{pr}_2(\phi(s_1(p)))$, such that for all $1 \leq j \leq k$ and all $1 \leq i_1, \dots, i_j \leq \dim(M)$:
$$\left.\frac{\partial^j(\psi \circ \text{pr}_2 \circ \phi \circ s_1 \circ \theta^{-1})}{\partial x^{i_1} \cdots \partial x^{i_j}}\right|_{\theta(p)} = \left.\frac{\partial^j(\psi \circ \text{pr}_2 \circ \phi \circ s_2 \circ \theta^{-1})}{\partial x^{i_1} \cdots \partial x^{i_j}}\right|_{\theta(p)}$$

In this case we write $s_1 \sim_p^k s_2$.

Prove: $\sim_p^k$ is an equivalence relation. Prove: If derivatives are equal for some chart, then it i s equal for all charts.

Definition 28.2.3 (Jet Equivalence)
Let $\pi : E \to M$ be a smooth fiber bundle with fiber F. Define the kth jet bundle of E to be the set

$$J^k(E) = \coprod_{p \in M} \frac{\{s_p : U \to E \mid s \text{ is a smooth local section containing } p\}}{\sim_p^k}$$

together with the following data.

- A smooth manifold structure given as follows. Fix an element $[s_p] \in J^k(M)$ for $p \in M$. Fix a chart (U, θ) at p, a trivialization chart (A, ϕ) of π, a chart (W, ψ) of F at $\text{proj}_F(\phi(s(p)))$, then a local chart of M at $[s_p]$ is given by

$$(\mathcal{U}, (\theta, u, u_{i_1 \cdots i_j} \mid 1 \leq j \leq k, 1 \leq i_1, \dots, i_j \leq \dim(M)))$$

where $\mathcal{U} = \{[t_p] \in J^k(M) \mid p \in U, t(p) \in \pi^{-1}(A), \text{pr}_F(\phi(s(p))) \in W\}$ and

$$u = (\psi \circ \text{proj}_F \circ \phi \circ s)(p) \in \mathbb{R}^{\dim(F)}$$

and

$$u_{i_1,\dots,i_j} = \frac{\partial^j(\psi \circ \text{proj}_F \circ \phi \circ s \circ \theta^{-1})}{\partial x^{i_j} \cdots \partial x^{i_1}}\bigg|_{\theta(p)}$$

- The map $J^k(E) \to M$ given by $[s_p] \to p$.

Lemma 28.2.4
Let $\pi : E \to M$ be a smooth fiber bundle with fiber F. Then the following are true.
- $J^k(E)$ is a smooth vector bundle over M.
- The manifold dimension of $J^k(E)$ is

$$\dim(J^k(E)) = \dim(M) + \dim(F) \sum_{j=0}^{k} \binom{\dim(M) + j - 1}{j}$$

- For each $p \in M$, there is a diffeomorphism

$$\pi^{-1}(p) \cong \mathbb{R}^{\dim(J^k(E)) - \dim(M)}$$

Definition 28.2.5 (Lagrangian Density)

Definition 28.2.6 (Lagrangian Function)

Definition 28.2.7 (Contact Forms)

Prove: A Lagrangian density modulo contact forms is a Lagrangian function tensored with a volume form.

Definition 28.2.8 (The Action Functional of a Lagrangian Density)

29 Smooth Principle Bundles

29.1 Basic Structures

Proposition 29.1.1
Let M be a smooth manifold. Let G be a Lie group with a free and smooth action on M. Then the following are true.
- For each $x \in M/G$, there exists a neighbourhood U of x and a map $s : U \to M$ such that $\pi \circ s = \text{id}_U$, where $\pi : M \to M/G$ is the projection map.
- The projection map $M \to M/G$ is a principal G-bundle.

Example 29.1.2
Let G be a Lie group. Let $H \leq G$ be a Lie subgroup. Then G acts on H by left multiplication. The above prp shows that G/H is a Lie group and $G \to G/H$ is a principal G-bundle since the orbit space of G acting on H is just G/H.

Theorem 29.1.3 (Poincaré-Hopf Theorem)
Let M be a smooth manifold. Then we have

$$e(TM)([M]) = \chi(M)$$

29.2 Connections on a Principal Bundle

Definition 29.2.1 (The Vertical Bundle)
Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \to M$ be a smooth principal bundle over G. Define the vertical bundle of P to be the smooth subbundle

$$VP = \coprod_{p \in P} \ker(d\pi_p)$$

Lemma 29.2.2
Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \to M$ be a smooth principal bundle over G. Let $p \in P$. Then there is an isomorphism

$$\iota_p : \text{Lie}(G) \stackrel{\cong}{\to} (VP)_p$$

given by $g \mapsto \frac{de^{tg} \cdot p}{dt}\bigg|_{t=0}$.

Definition 29.2.3 (Horizontal Distributions)
Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \to M$ be a smooth principal bundle over G. A horizontal distribution on P is a subbundle $HP \subseteq TP$ such that the following are true.
- $TP = VP \oplus HP$.
- $(HP)_{g \cdot p} = (L_g)_*(HP)_p$ for all $p \in P$ and $g \in G$.

Generalizes connections on vector bundles in the following way: For $p : E \to M$ a smooth vector bundle, a connection on $\text{Fr}(E)$ as a principal $\text{GL}(n, F)$ bundle is the same data as a connection on $p : E \to M$ as a vector bundle.

Definition 29.2.4 (Connection Forms on Principal Bundles)
Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \to M$ be a smooth principal bundle over G. A connection 1-form on P is a smooth section $\omega \in \mathcal{C}^\infty(T^*P \otimes \text{Lie}(G))$ such that the following are true.
- We have $\omega_p(\iota_p) = \text{id}_{\text{Lie}(G)}$ for all $p \in P$.

- We have
$$\omega_{g\cdot p}((dL_g)_p(v)) = \text{Ad}(g)(\omega_p(v))$$
for all $g \in G$, $p \in P$ and $v \in T_pP$.

Proposition 29.2.5
Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \to M$ be a smooth principal bundle over G. Then the following are true.
- Suppose that HP is a horizontal distribution on P. Define a 1-form ω pointwise by $\omega_p(X) = \iota_p^{-1}(X^V)$ where X^V is the vertical component of X in the decomposition $T_pP = (VP)_p \oplus (HP)_p$. Then ω is a connection 1-form.
- Suppose that ω is a connection 1-form on P. Then $\coprod_{p\in P} \ker(\omega_p)$ is a horizontal distribution on P.

Definition 29.2.6 (The Curvature Form of a Connection)
Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \to M$ be a smooth principal bundle over G. Let ω be a connection 1-form. Define the curvature form of ω to be the 2-form $\Omega \in \mathcal{C}^\infty(\Lambda^2 T^*P \otimes \text{Lie}(G))$ given by
$$\Omega = d\omega + \frac{1}{2}[\omega \wedge \omega]$$

This definition is motivated by the local description of the curvature form for vector bundles.

Using tensor algebra, we see that $T^*P \otimes T^*P \otimes \mathfrak{g} \cong TP \times TP \to (P \times \mathfrak{g})$. Since $\Lambda^2 T^*P$ is the subbundle of $T^*P \otimes T^*P$ consisting of alternating forms, we may also think of $\Omega \in \mathcal{C}^\infty(\Lambda^2 T^*P \otimes \mathfrak{g})$ as a map $TP \times TP \to P \times \mathfrak{g}$ satisfying $\Omega(X, X) = 0$. Also, for each $p \in P$, $\Omega_p : T_pP \times T_pP \to \mathfrak{g}$ is then an alternating bilinear map.

Definition 29.2.7 (Components of the Curvature Form)
Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \to M$ be a smooth principal bundle over G. Let ω be a connection 1-form. Let Ω be the curvature form. Let $e_1, \dots, e_n$ be a basis of $\mathfrak{g}$. Define the components of Ω to be the 2-forms $\Omega^i \in \Omega^2(P)$ such that
$$\Omega = \sum_{i=1}^n \Omega^i \otimes e_i$$

29.3 The Chern-Weil Homomorphism

Let V be a finite dimensional vector space over a field k. Recall from Rings and Modules that there is an isomorphism of graded F-algebras $S(V) \cong k[x_1, \dots, x_n]$.

Definition 29.3.1 (Ad-Invariant Polynomials)
Let G be a Lie group. Let $f \in S(\text{Lie}(G)^*)$. We say that f is ad-invariant if $f(\text{Ad}_g(X)) = X$ for all $X \in \text{Lie}(G)$.

Definition 29.3.2 (The Algebra of Ad-Invariant Polynomials)
Let G be a Lie group. Define $\mathbb{R}$-algebra of ad-invariant polynomials of G by
$$I(G) = \{f \in S(\text{Lie}(G)^*) \mid f \text{ is ad-invariant }\}$$

Let $f \in S(\text{Lie}(G)^*)$ be a polynomial. Let Ω be a curvature 2-form. For each $p \in P$, we know that Ω_p is an alternating bilinear map from T_pP to $\text{Lie}(G)$.

Definition 29.3.3 (Evaluating Polynomials on the Curvature)
Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \to M$ be a smooth principal bundle over G. Let ω be a connection 1-form. Let $f \in I(G)$ be an ad-invariant polynomial of the form $f = \sum_{i_1,\dots,i_n} \alpha_{i_1,\dots,i_n} x_1^{i_1} \cdots x_n^{i_n}$. Let Ω be the curvature 2-form of ω. Define the evaluation of f on Ω to be the $2k$-form $f(\Omega) \in \Omega^{2k}(M)$ given by

$$f(\Omega) = \sum_{i_1,\dots,i_n} \alpha_{i_1,\dots,i_n} (\Omega^1)^{\wedge i_1} \wedge \cdots \wedge (\Omega^n)^{i_n}$$

Lemma 29.3.4
Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \to M$ be a smooth principal bundle over G. Let $f \in I(G)$ be an ad-invariant polynomial. Let ω be a connection 1-form. Let Ω be the curvature form. Then we have

$$df(\Omega) = 0$$

Lemma 29.3.5
Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \to M$ be a smooth principal bundle over G. Let $f \in I(G)$ be an ad-invariant polynomial. Let ω, ω' be connection 1-forms. Let Ω and Ω' be the curvature forms respectively. Then we have

$$[f(\Omega)] = [f(\Omega')] \in H^{2k}_{\text{dR}}(M)$$

Definition 29.3.6 (The Chern-Weil Homomorphism)
Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \to M$ be a smooth principal bundle over G. Let Ω be the curvature form of any connection. Define the Chern-Weil homomorphism to be the map

$$P(\Omega) : S(\text{Lie}(G)^*) \to H^*_{\text{dR}}(M)$$

given by $f \mapsto [f(\Omega)]$.

Lemma 29.3.7
Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \to M$ be a smooth principal bundle over G. Let Ω be the curvature form of any connection. Then $P(\Omega)$ is an algebra homomorphism.

Proposition 29.3.8
Let M be a smooth manifold. Let $\pi : P \to M$ be a smooth vector bundle of dimension n over $\mathbb{C}$. Let Ω be the curvature form of any connection. Suppose that $f_0, \dots, f_n \in S(\mathfrak{gl}(n, \mathbb{C})^*)$ are such that $c_X(\lambda) = \sum_{i=0}^n f_i(X)\lambda^{n-i}$ Then the following are true.
- $f_0, \dots, f_n$ generate $S(\mathfrak{gl}(n, \mathbb{C})^*)$ as a $\mathbb{C}$-algebra.
- $[f_1(\Omega)], \dots, [f_n(\Omega)]$ are the Chern classes of P.

Proposition 29.3.9
Let M be a smooth manifold. Let $\pi : P \to M$ be a smooth vector bundle of dimension n over $\mathbb{R}$. Let Ω be the curvature form of any connection. Suppose that $f_0, \dots, f_n \in S(\mathfrak{gl}(n, \mathbb{R})^*)$ are such that $c_X(\lambda) = \sum_{i=0}^n f_i(X)\lambda^{n-i}$ Then the following are true.
- $f_0, \dots, f_n$ generate $S(\mathfrak{gl}(n, \mathbb{R})^*)$ as a $\mathbb{R}$-algebra.
- $[f_1(\Omega)], \dots, [f_n(\Omega)]$ are the Pontrjagin classes of P.

30 The Logical Chain of Classical Field Theory and Gauge Theory

Step 1: The Field

We observe a physical quantity distributed over spacetime — a temperature, a velocity, an electromagnetic strength. Mathematically this is a fiber bundle $\pi : E \to M$, where M is spacetime and the fiber E_x is the space of possible values of the quantity at x. A **section** $\phi \in \Gamma(E)$ is one complete specification of the quantity at every point — a possible physical configuration.

Step 2: The Lagrangian as a Choice of Physics

A Lagrangian density $\Lambda \in \Omega^n(J^k E)$ is a choice of action functional

$$S[\phi] = \int_\Omega (j^k\phi)^*\Lambda$$

Different Lagrangians encode different physical theories. The Euler-Lagrange equations — the equations of motion — are derived from Λ by the variational principle. The physical sections are the critical points of S.

Step 3: Observing a Symmetry

You compute the Euler-Lagrange equations and notice that the Lagrangian Λ is invariant under some subgroup of automorphisms of E. An **automorphism** of $\pi : E \to M$ is a fiber-preserving diffeomorphism $\Phi : E \to E$. Such an automorphism acts on sections by

$$\phi \mapsto \Phi \circ \phi$$

and the Lagrangian being invariant means

$$S[\Phi \circ \phi] = S[\phi] \quad \text{for all } \phi \in \Gamma(E)$$

At this stage, the symmetry group acts the same way at every point — it is **global**. You observe this as an empirical fact about your Lagrangian.

Symmetries of the action induces certain laws of physics, such as conservation of energy, momentum, charge etc. This is Noether's theorem.

Another note: Rather than starting with a Lagrangian and observe the symmetries of its action, we may also start with a set of symmetries that we would like to enforce, apply it to all differential forms and retain only the invariants, and see what Lagrangian you would get.

Step 4: The Absolute vs Relative Observation

The global symmetry exists because the field takes values in a space that has **no preferred origin or frame** — the fiber E_x has a group of self-maps (rotations, phase shifts, color rotations) that are physically unobservable. Different observers, or different conventions, would describe the same physical situation using sections related by this group action. The global symmetry is an artifact of having made consistent conventional choices everywhere at once. We need not restrict ourselves to consider symmetries of the entire space, and this is supported by physics:

Argument 1: Special Relativity

Special relativity says there is no notion of simultaneity — two spacelike separated points have no canonical notion of "the same moment." A global transformation is therefore not a physically coherent operation in a relativistic theory. If you believe in special relativity, you are forced to allow your symmetry to act differently at different points. The global symmetry is then a special case of the local one, not the fundamental object.

Argument 2: The Gauge Principle

The difference induced by the symmetry of the action is not observable. The phase is therefore a **convention** — a choice made by an observer at x to describe the field. If the phase is a convention, then different observers at different points of spacetime should be free to make different conventional choices independently. There is no physical mechanism that would force their conventions to agree globally. The true symmetry group of the theory is therefore not the global group G but the local group $mathcalG = \mathrm{Map}(M, G)$ of all smooth G-valued functions on spacetime.

This is the **gauge principle**: any redundancy in the description of a physical system should be a local redundancy. It is a postulate, not a theorem. Its justification is that it works — every time it is applied, it produces a physically correct and predictive theory.

Step 5: Promoting to Local Symmetry

This forces you to demand invariance not just under global G-transformations

$$\phi(x) \mapsto g \cdot \phi(x), \quad g \in G \text{ constant}$$

but under local G-transformations

$$\phi(x) \mapsto g(x) \cdot \phi(x), \quad g : M \to G \text{ smooth}$$

The Lagrangian, built from ordinary derivatives $\partial_\mu \phi$, immediately fails to be locally invariant. The reason is fundamental: $\partial_\mu \phi$ compares field values at nearby points x and $x + dx$, which under local symmetry live in fibers with **independent conventions**. Subtracting them is meaningless — it is comparing quantities measured in different units.

Step 6: The Mathematics Is Now Forced

To make sense of differentiation under local symmetry, we need a way to **compare vectors in nearby fibers** compatibly with the G-structure. But this is precisely the definition of a **connection**. Where does this connection live? The answer is that the principal bundle is not introduced from outside — it is **already contained inside the vector bundle** E.

The fibers E_x are vector spaces, but the isomorphism $E_x \cong \mathbb{R}^r$ is not canonical — it requires a choice of basis. A **frame** at x is an ordered basis for E_x. The set of all frames at x carries a natural free transitive right action of $GL(r, \mathbb{R})$, making it a $GL(r, \mathbb{R})$-torsor. Taking the disjoint union over all $x \in M$ produces the **frame bundle**

$$P := \bigsqcup_{x \in M} \{\text{ordered bases for } E_x\}$$

a principal $GL(r, \mathbb{R})$-bundle canonically constructed from E.

Now, the observation of a global G-symmetry in Step 3 is precisely the observation that the physically relevant frames are not arbitrary frames, but only those related by G-transformations. The structure group **reduces** from $GL(r, \mathbb{R})$ to G, producing a principal G-bundle

$$P_G \subset P$$

canonically determined by the symmetry. The original vector bundle is recovered as the associated bundle $E = P_G \times_G V$.

The connection demanded by local symmetry lives on P_G, because P_G is the bundle whose fibers are G-torsors — they encode exactly the gauge freedom at each point. A connection on P_G transports G-frames between nearby fibers, and therefore transports everything associated to those frames, including sections of $E = P_G \times_G V$.

Step 7: What Follows From The Maths

Since the principal bundle is just a collection of frames at each fiber. A local section of the principal bundle is a choice of local coordinates. The mathematical fact that non-trivial principal bundles have no global sections can be interpreted as the fact that there is generally no globally consistent choice of coordinates because the symmetries now act locally.

A connection in the principal bundle gives a covariant derivative in the associated bundle. This gives a way to compare phases across different time space.

31 Intersection Theory

31.1 The Divisor of a Rational Function

Definition 31.1.1 (The Order of a Function at a Subvariety)
Let k be a field. Let X be a scheme of finite type over k. Let W be an irreducible subvariety of X. Let $f/g \in K(W)^*$ for $f, g \in \mathcal{O}_X(W)$ and $g \neq 0$. Let $V \subseteq W$ be an irreducible subvariety of codimension 1. Define the order of f at V to be

$$\text{ord}_V(f/g) = l_{\mathcal{O}_X(W)}\left(\frac{\mathcal{O}_X(W)}{(f)}\right) - l_{\mathcal{O}_X(W)}\left(\frac{\mathcal{O}_X(W)}{(g)}\right)$$

Definition 31.1.2 (The Divisor of a Rational Function)
Let k be a field. Let X be a scheme of finite type over k. Let W be an irreducible subvariety of X. Let $f \in K(W)^*$. Define the divisor of f to be

$$\text{div}(f) = \sum_{\substack{V \text{ is a codimension 1} \\ \text{irreducible subvariety of } W}} \text{ord}_V(f) \cdot V$$

31.2 The Chow Groups

Definition 31.2.1 (The Group of n-Cycles)
Let k be a field. Let X be a scheme of finite type over k. Let $n \in \mathbb{N}$. Define the group of n-cycles on X to be the free abelian group

$$Z_n(X) = \mathbb{Z}\langle V \mid {}_{\text{irreducible subvariety of } X}^{V \text{ is an } n\text{-dimensional}} \rangle$$

Lemma 31.2.2
Let k be a field. Let X be a scheme of finite type over k. Let W be an irreducible subvariety of X. Let $f \in K(W)^*$. Then $\text{div}(f) \in Z_{\dim(W)-1}(X)$.

Definition 31.2.3 (Cycles Rationally Equivalent to Zero)
Let k be a field. Let X be a scheme of finite type over k. Let $V \in Z_n(X)$. We say that V is rationally equivalent to 0 if there exists finitely many $n+1$-dimensional irreducible subvarieties $W_1, \dots, W_r$ of X and $f_i \in K(W_i)^*$ such that

$$V = \sum_{i=1}^n \text{div}(f_i)$$

Proposition 31.2.4
Let k be a field. Let X be a scheme of finite type over k. Let $V \in Z_n(X)$ be a cycle. Then V is rationally equivalent to 0 i and only if there exists $(n+1)$-dimensional irreducible subvarieties $W_1, \dots, W_r$ of $X \times \mathbb{P}^1_k$ such that the projection maps $p_i : W_i \to \mathbb{P}^1_k$ are dominant and $V = \sum_{i=1}^r [p_i^{-1}(0)] - [p_i^{-1}(\infty)]$

Definition 31.2.5 (The Chow Group)
Let k be a field. Let X be a scheme of finite type over k. Define the nth Chow group of X to be

$$\text{CH}_n(X) = \frac{Z_n(X)}{\sim}$$

where we say that $V_1 \sim V_2$ if $V_1 - V_2$ is rationally equivalent to 0. In this case, we say that V_1 and V_2 are rationally equivalent.

31.3 The Ring Structure

31.4 Pushfoward and Pullback of Cycles

Definition 31.4.1 (The Push Forward Map)
Let k be a field. Let X, Y be schemes of finite type over k. Define the push forward map $f_* : Z_n(X) \to Z_n(Y)$ by

$$f_*(V) = \deg(V/f(V))f(V)$$

where

$$\deg(V/f(V)) = \begin{cases} [k(V) : k(f(V))] & \text{if } \dim(V) = \dim(f(V)) \\ 0 & \text{otherwise} \end{cases}$$

Proposition 31.4.2
Let k be a field. Let X, Y be schemes of finite type over k. Then the push forward map descends to a well defined homomorphism $f_* : \text{CH}_n(X) \to \text{CH}_n(Y)$.

Proposition 31.4.3
Let k be a field. Let X be a scheme of finite type over k. Let $i : Y \hookrightarrow X$ be a closed subscheme of X. Let $j : X \setminus Y \hookrightarrow X$ be the inclusion map. Then there is an exact sequence

$$\text{CH}_n(Y) \xrightarrow{i_*} \text{CH}_n(X) \xrightarrow{j^*} \text{CH}_n(X \setminus Y) \longrightarrow 0$$

32 Algebraic de Rham Cohomology

Definition 32.0.1 (Algebraic de Rham Cohomology)
Let S be a scheme. Let X be a scheme over S. Define the nth algebraic de Rham cohomology group by

$$H^n_{\text{dR}}(X/S) = \mathbb{H}^n(X; \Omega^\bullet_{X/S})$$

Lemma 32.0.2
Let S be a scheme. Let X be a scheme over S. Then $\text{Tot}^{\oplus}\left(\mathcal{C}^\bullet(X, \mathcal{U}, \Omega^\bullet_{X/S})\right)$ is an acyclic resolution of $\Omega^\bullet_{X/S}$.

Thus one way to compute the de Rham cohomology would be

$$H^n_{\text{dR}}(X/S) \cong H^n(\text{Tot}^{\oplus}(\mathcal{C}^\bullet(X, \mathcal{U}, \Omega^\bullet_{X/S}))$$

Example 32.0.3
Let $U_0 = \text{Spec}(R_0 = k[x, y]/(y^2 - f(x)))$ where k is a field of characteristic 0 and f is separable and $\deg(f) = 2g+1$ or $2g+2$. Let $U_1 = \text{Spec}(R_1 = k[u, v]/(v^2 - h(u)))$ where $h(u) = u^{2g+2}h(1/u)$. Then we have

$$H^i_{\text{dR}}(X/k) = H^i\left(R_0 \oplus R_1 \to R_0 \frac{dx}{y} \oplus R_1 \frac{du}{v} \oplus R_0[1/x] \to R_0[1/x]dx\right)$$

where the first map is given by $(w, z) \mapsto (dw, dz, w - z(1/x, y/x^{g+1}))$, the second map is given by $(wdx, zdu, g(x, y)) \mapsto wdx - zdu - dg$ where $du = -z(1/x, y/x^{g+1})dx/x^2$.

- $H^0_{\text{dR}}(X/k) \cong k$.
- $H^2_{\text{dR}}(X/k) \cong H^1_{\text{dR}}(X/k)$.
- $H^0(X, \Omega^0_{X/k}) \hookrightarrow H^1_{\text{dR}}(X/k)$ given by $\omega \mapsto (\omega|_{U_0}, \omega|_{U_1}, 0)$ is an inclusion since the global sections are constant.
- There is a map $H^1_{\text{dR}}(X/kk) \to H^1(X, \mathcal{O}_X$ given by $(\omega_0, \omega_1, g) \mapsto g$ and precomposing with the above map gives 0. It is clearly surjective.

- So we have an exact sequence
$$0 \to H^0(X, \Omega^\bullet_{X/k}) \to H^1_{\mathrm{dR}}(X/k) \to H^1(X, \mathcal{O}_X) \to 0$$
called the Hodge filtration on $H^1_{\mathrm{dR}}(X/k)$. In characteristic p this is more subtle: not all global differentials are closed.
- What can we say about the map $H^1_{\mathrm{dR}}(X/k) \to H^1_{\mathrm{dR}}(U_0/k)$ given by $(\omega_0, \omega_1, g) \mapsto g$.
- The LES in local cohomology gives
$$0 \to H^1(X(\mathbb{C}), \mathbb{C}) \to H^1(U_0(\mathbb{C}), \mathbb{C}) \to H^2(X(\mathbb{C}), U_0(\mathbb{C}), \mathbb{C}) \to H^2(X(\mathbb{C}), \mathbb{C}) \to 0$$
and $H^2(X(\mathbb{C}), U_0(\mathbb{C}), \mathbb{C}) \cong H^0((X - U_0)(\mathbb{C}), \mathbb{C}) \cong \mathbb{C}^{|X - U_0|(\mathbb{C})}$. If $\deg(f)$ is odd, $|(X - U_0)(\mathbb{C})| = 1$. Then $H^1(X(\mathbb{C}), \mathbb{C}) \to H^1(U_0(\mathbb{C}), \mathbb{C})$ is an isomrophism. If $\deg(f)$ is even, then we have $0 \to H^1(X(\mathbb{C}), \mathbb{C}) \to H^1(U_0(\mathbb{C}), \mathbb{C}) \to \mathbb{C} \to 0$. Same thing happens for de Rham cohomology. Surjectivity of $H^1_{\mathrm{dR}}(X/k) \to H^1_{\mathrm{dR}}(U_0/k)$ iff every class (ω_0) for $\omega_0 \in \Omega^1_{R_0/k}$ is the image of $\ker(R_0 dx/y \oplus R_1 du/v \oplus R_0[1/x] \to R_0[1/x]dx)$. i.e. $\exists \omega_1 \in R_1 du/v$, $g \in R_0[1/x]$ such that $\omega_0 - \omega_1 = dg$. The obstruction to finding such an ω_1 and g would be the residue of a differential. Pick $z \in (X - U_0)(\overline{k})$ a closed point and a parameter t at z. Write ω as $F(t) = \sum_{i=0}^\infty a_i t^i \in \overline{k}[[t]]$. Eg any meromorphic can be written as rational function times dt, then, take formal power series. Then $\mathrm{Res}_z(\omega) = a_1$ of $F(t)$. If $\omega_1 \in \Omega^1_{R_1/k}$ then $\mathrm{Res}_z = 0$ (indeed it has no pole). Also $\mathrm{Res}_z(dg) = 0$. Then image of $H^1_{\mathrm{dR}}(X/k)$ is contained in the kernel of the residue map
$$\mathrm{Res} : H^0_{\mathrm{dR}}(U_0/k) \to \bigoplus_{Z \in (X - U_0(\overline{k})} \overline{k}$$
Fact: For any open $U \subseteq X$, $\omega \in H^0(U, \Omega_{X/k})$, then
$$\sum_{Z \in (X-U)(\overline{k})} \mathrm{Res}_z(\omega) = 0$$

Proposition 32.0.4
Let S be a scheme. Let X be a scheme over S. Then the following are true.
- $H^n_{\mathrm{dR}}(X/S)$ is a $H^0_{\mathrm{dR}}(X/S)$-module.
- The map $H^i_{\mathrm{dR}}(X/S) \times H^j_{\mathrm{dR}}(X/S) \to H^{i+j}_{\mathrm{dR}}(X/S)$ induced by the wedge of differential forms is a $H^0_{\mathrm{dR}}(X/S)$-linear.

Definition 32.0.5 (The de Rham Cohomology Ring)
Let S be a scheme. Let X be a scheme over S. Define the de Rham cohomology ring of X to be
$$H^*_{\mathrm{dR}}(X/S) = \bigoplus_{i=0}^\infty H^i_{\mathrm{dR}}(X/S)$$
together with ring structure given by wedge of differential forms.

Lemma 32.0.6
Let S be a scheme. Let X be a scheme over S. Then $H^*_{\mathrm{dR}}(X/S)$ is a graded commutative $H^0_{\mathrm{dR}}(X/S)$-algebra.

Lemma 32.0.7
Let k be a field of characteristic 0. Let X be a smooth affine variety over k. Then we have
$$H^n_{\mathrm{dR}}(X) = H^n(\Gamma(X, \Omega^\bullet_{X/k}))$$
for all $n \in \mathbb{N}$.

Example 32.0.8
Let k be a field of characteristic 0. Let $R = k[x_1, \dots, x_n]$ be the polynomial ring over k. Then $\Gamma\left(X, \Omega^1_{\text{Spec}(R)/k}\right) = \Omega^1_{R/k} = \bigoplus_{i=1}^n R dx_i$. Then we have

$$\Gamma\left(X, \Omega^j_{\text{Spec}(R)/k}\right) = \bigoplus_{1 \leq i_1 < \cdots < i_j \leq n} R dx_{i_1} \wedge \cdots \wedge dx_{i_j}$$

Then we have

$$H^i_{\text{dR}}(R/k) = \begin{cases} k & \text{if } i = 0 \\ 0 & \text{otherwise} \end{cases}$$

TBR:

Example 32.0.9
Let k be a field of characteristic $p > 0$. Then we have

$$\dim_k(H^1(\Gamma(\text{Spec}(k[t]), \Omega^1_{\text{Spec}(k[t])/k}))) = \infty$$

Indeed, we have

$$H^1(\Gamma(\text{Spec}(k[t]), \Omega^1_{\text{Spec}(k[t])/k})) = \frac{k[t]}{\langle f'(t) \mid f \in k[t] \rangle} = \frac{k[t]}{\bigoplus_{i=1}^\infty i t^{i-1} dt} \cong \bigoplus_{i=1}^\infty t^{ip-1} dt$$

Example 32.0.10
$\Omega^1_{\mathbb{Z}[\sqrt{2}]/\mathbb{Z}} \cong (\mathbb{Z}/2\mathbb{Z}) d\sqrt{2}$. $H^1(\Gamma(\text{Spec}(\mathbb{Z}\sqrt{2}), \Omega^1_{\mathbb{Z}[\sqrt{2}]/\mathbb{Z}})) \neq 0$

Example 32.0.11
Let k be a field of characteristic $p \neq 2$. Let $f \in k[x]$ be separable. Let $R = \frac{k[x,y]}{y^2 - f(x)}$. Then we have

$$\Omega^1_{R/k} \cong \frac{\Omega^1_{k[x,y]/k} \otimes_{k[x,y]} k}{(2ydy - f'(x)dx)} \cong \frac{Rdx \oplus Rdy}{R(2ydy - f'(x)dx)}$$

Using the quotient relation, we may write $dy = \frac{f'(x)dx}{2y}$ in the module. Thus dx/y is a generator of the module.

Since f is separable, there exists $g, h \in k[x]$ such that $gf + hf' = 1$. Let $\omega = aydx + 2bdy \in \Omega^1_{R/k}$. Then we have $y\omega = dx$. Hence y divides dx and $\omega = dx/y$.

Now let $z = a(x) + b(x)y \in R$ be arbitrary. Then we have

$$dz = (a'(x) + b'(x)y)dx + h(x)dy = (ya'(x) + b'(x)f(x) + \frac{1}{2}b(x)f'(x))\omega$$

Thus every element of $\Omega^1_{R/k}$ is of the form $q(x)\omega$ for some $q(x) \in k[x]$. If $\deg(q) \geq \deg(f)$, then there exists $p \in k[x]$ such that $\deg(p) > \deg(q)$ such that

$$[p(x)\omega] = [q(x)\omega]$$

in $H^1_{\text{dR}}(R/k)$.

Lemma 32.0.12
Let k be a field of characteristic 0. Let X be a smooth variety over k. Then the following are true.
- $H^i_{\text{dR}}(X) = 0$ for all $i > 2\dim(X)$.
- If X is affine, then $H^i_{\text{dR}}(X) = 0$ for all $i > \dim(X)$.

Proposition 32.0.13

32.1 Hodge de Rham Spectral Sequences

Definition 32.1.1 (The Hodge Filtration)
Let S be a scheme. Let X be a scheme over S. Define the Hodge filtration $F^\bullet\Omega^\bullet_{X/S}$ of X to be

$$F^i\Omega^j_{X/S} = \begin{cases} \Omega^j_{X/S} & \text{if } j \geq i \\ 0 & \text{otherwise} \end{cases}$$

Thus for each i, the collection of complexes

$$F^i\Omega^\bullet_{X/S} = (0 \to \Omega^i_{X/S} \to \Omega^{i+1}_{X/S} \to \cdots)$$

form the filtration for $\Omega^\bullet_{X/S}$.

Lemma 32.1.2
Let S be a scheme. Let X be a scheme over S. Then $F^\bullet\Omega^\bullet_{X/S}$ is an exhaustive filtration.

Definition 32.1.3 (The Hodge de Rham Spectral Sequence)
Let S be a scheme. Let X be a scheme over S. Define the Hodge de Rham spectral sequence to be the spectral sequence associated to the Hodge siltration $F^\bullet\Omega^\bullet_{X/S}$.

The E^0 page is given by

$$E_0^{p,q} = \frac{F^p\Omega^{p+q}_{X/S}}{F^{p+1}\Omega^{p+q}_{X/S}} = \frac{\Omega^q_{X/S}}{\Omega^{q-1}_{X/S}}$$

Lemma 32.1.4
Let S be a scheme. Let X be a scheme over S. Then we have

$$E_1^{p,q} = H^q(X, \Omega^p_{X/S}) \Rightarrow H^{p+q}_{\text{dR}}(X/S)$$

Theorem 32.1.5 (Deligne)
If $S = \text{Spec}(k)$, k char 0, $X \to S$ smooth and proper. Then the Hodge de Rham spectral sequence degenerates at the E_1 page (all differentials in page 1 is 0).

When you have a 2-step filtration: $0 \to F^1C^\bullet \to C^\bullet \to C^\bullet/F^1C^\bullet \to 0$, the spectral sequence is the long exact sequence in cohomology.

32.2 Flat Connections

$X/\mathbb{C}$ smooth connected algebraic variety. $(V, \Delta) \to (V^{\text{an}})^{\Delta=0}$.

Cyclic vector theorem: (V, Δ) on $K(C)$, C a connected curve over k. Suppose that $e \in V$. Let $e_i = \partial_i e$.

33 Deformation of Manifold Structures

33.1

Recall:

$$\begin{tikzcd}
\text{Smooth Manifolds} \arrow[r, phantom, "\subset"] & \text{PL Manifolds} \arrow[r, phantom, "\subset"] \arrow[d, leftrightarrow] & \text{Topological Manifolds} \\
& \text{Combinatorial Manifolds} \arrow[r, phantom, "\subset"] & \text{Triangulated Manifolds} \arrow[u]
\end{tikzcd}$$

is every triangulated manifold a combinatorial manifold? manifold Hauptvermutung:

Definition 33.1.1 (Smooth Embedding of a Simplex)
Let $\Delta^k \subseteq \mathbb{R}^{k+1}$ be the standard k-simplex. Let M be a smooth manifold. Let $\sigma : \Delta^k \to M$ be a map. We say that σ is a smooth embedding if the following are true.

- f is injective.
- f induces a homeomorphism from Δ^k to $\text{im}(f)$.
- For all facets $F \subseteq \Delta^k$, the induced map $\sigma|_{F^\circ}$ is an immersion of smooth manifolds.

Definition 33.1.2 (Smooth Triangulations)
Let M be a smooth manifold with boundary. Let K be a simplicial complex that is a triangulation of M. We say that K is a smooth triangulation if the following are true.

- For each simplex $\sigma \in K$, the map σ is a smooth embedding.
- There exists a subcomplex $K' \subseteq K$ such that K' is a triangulation of ∂M.

Lemma 33.1.3
Let M be a smooth manifold with boundary. Let K be a smooth triangulation of M. Then M admits a PL structure.

Theorem 33.1.4 (Whitehead, 1940)
Let M be a smooth manifold with boundary. Then M admits a smooth triangulation. Moreover, if K and K' are both smooth triangulations, then $|K|$ and $|K'|$ are PL-homeomorphic.

33.2 Microbundles

Define the tangent microbundle τ_M for a topological n-manifold M. Stable topological microbundles over M are classified by $BTop = \varinjlim_k B\text{Top}(k)$, so the stable tangent microbundle determines a classifying map $M \to BTop$. Since $\text{Top}(k)/\text{PL}(k) \to Top/PL$ is a homotopy equivalence for $k \geq 5$, a lift of $M \to BTop$ through $BPL \to BTop$ is equivalent to a PL structure on the unstable tangent microbundle τ_M.

Lemma 33.2.1
The following are true.

- There is a homotopy equivalence
$$\text{hofib}(BPL \to BTop) \simeq K(\mathbb{Z}/2\mathbb{Z}, 3)$$
- $BPL \to BTop$ is a principal fibration.
- The principal fibration induces a fibration $BTop \to K(\mathbb{Z}/2\mathbb{Z}, 4)$.

33.3 The Tangent Microbundle

Theorem 33.3.1 (Hirsch-Mazur Theorem)
Let M be a topological manifold of dimension $n \geq 5$. Then τ_M admits a PL structure if and only if M admits a PL structure.

33.4 The Kirby-Siebenmann Class

Definition 33.4.1 (The Kirby-Siebenmann Class)
Let M be a topological manifold. Define the Kirby-Siebenmann class of M to be the element

$$\kappa(M) \in H^4(M; \mathbb{Z}/2\mathbb{Z}) \cong [M, K(\mathbb{Z}/2\mathbb{Z}, 4)]$$

given by the element $M \overset{\tau M}{\to} BTop \to K(\mathbb{Z}/2\mathbb{Z}, 4)$.

Proposition 33.4.2 (Kirby-Siebenmann Structure Theorem)
Let M be a compact topological manifold of dimension $n \geq 5$. Then the following are true.
- M admits a PL structure if and only if $\kappa(M) = 0$.
- If $\kappa(M) = 0$, then the set of concordance classes of PL structure on M is a torsor over $H^3(M; \mathbb{Z}/2\mathbb{Z})$.

33.5 The Low Dimensional Case

Theorem 33.5.1 (Rado's Theorem)
Let M be topological manifold of dimension 2. Then M admits a unique smooth structure, unique PL structure and unique triangulation.

Theorem 33.5.2 (Moise's Theorem)
Let M be topological manifold of dimension 3. Then M admits a unique smooth structure, unique PL structure and unique triangulation.

Theorem 33.5.3 (Cairns-Hirsch)
Let M be piecewise linear manifold of dimension 4. Then M admits a unique smooth structure and a unique PL structure.

Theorem 33.5.4
Let M be a topological manifold of dimension 4. Then the following are true.
- If $\kappa(M) \neq 0$, then M admits no PL structure and no smooth structure.
- If $\kappa(M) = 0$ and Q_M is not diagonalizable over $\mathbb{Z}$, then M admits no PL structure and no smooth structure.

Theorem 33.5.5
The following are true.
- There exists uncountably many pairwise non-diffeomorphic smooth structures on $\mathbb{R}^4$.
- For $n \neq 4$, there exists a unique smooth structure on $\mathbb{R}^n$.

Theorem 33.5.6 (Manolescu 2016)
For all $n \geq 5$, there exists a compact topological manifold of dimension n that admits no triangulation.

Theorem 33.5.7
Let M be PL manifold of dimension $n \leq 6$. Then M admits a smooth structure.

$$\underset{\text{Manifolds}}{\text{Smooth}} \subset \underset{\text{Manifolds}}{\text{PL}} \subset \underset{\text{Manifolds}}{\text{Topological}}$$

Summarizing: All inclusions are equalities for $\dim \leq 3$. The first inclusion is strict for $\dim = 4$.

34 Rational Homotopy Theory

34.1 Homology with Coefficients in a Subring of $\mathbb{Q}$

Theorem 34.1.1 (Whitehead-Serre Theorem)
Let X, Y be simply connected spaces. Let R be a subring of $\mathbb{Q}$. Let $f : X \to Y$ be a map. Then the following are equivalent.
- $\pi_*(f) \otimes_{\mathbb{Z}} R$ is an isomorphism of graded abelian groups.
- $H_*(f; R)$ is an isomorphism of graded rings.
- $H_*(\Omega f; R)$ is an isomorphism of graded rings.

Lemma 34.1.2
Let X be an n-connected space. Let R be a subring of $\mathbb{Q}$. Suppose that one of the following are true.
- $n \geq 1$.
- $n = 0$ and $\pi_1(X)$ is abelian.

Then the Hurewicz homomorphism induces an isomorphism

$$\pi_{n+1}(X) \otimes_{\mathbb{Z}} R \cong H_{n+1}(X; R)$$

Proposition 34.1.3
Let A be an abelian group. Let $n \in \mathbb{N}^+$. Let R be a subring of $\mathbb{Q}$. Then we have $A \otimes_{\mathbb{Z}} R = 0$ if and only if $H_*(K(A, n); R) \cong H_*(*; R)$.

34.2 P-Local Modules

Definition 34.2.1 (P-Local Groups)
Let A be an abelian group. Let $P \subseteq \mathbb{N}$ be a set of prime numbers. We say that A is P-local if for all $q \in \mathbb{N} \setminus P$, the map $\cdot q : A \to A$ is an isomorphism.

Definition 34.2.2 (P-Localization Map)
Let A be an abelian group. Let $P \subseteq \mathbb{N}$ be a set of prime numbers. Define the P-localization map of A to be the map

$$A \to A \otimes_{\mathbb{Z}} S_P^{-1}\mathbb{Z}$$

where $S_P = \{n \in \mathbb{Z} \mid \gcd(n, p) = 1 \text{ for all } p \in P\}$.

Proposition 34.2.3
Let A be an abelian group. Let $P \subseteq \mathbb{N}$ be a set of prime numbers. Let $S_P = \{n \in \mathbb{Z} \mid \gcd(n, p) = 1 \text{ for all } p \in P\}$. Then the following are equivalent.
- A is P-local.
- A is a $S_P^{-1}\mathbb{Z}$-module by extension of scalars.
- The P-localization map $A \to A \otimes_{\mathbb{Z}} S_P^{-1}\mathbb{Z}$ is an isomorphism.

When $P = \emptyset$, then $S_P^{-1}\mathbb{Z} = \mathbb{Q}$ is just rationalization.

34.3 P-Local Spaces

Definition 34.3.1 (P-Local Space)
Let X be a simply connected space. Let $P \subseteq \mathbb{N}$ be a set of prime numbers. We say that X is P-local if $\pi_n(X)$ is a P-local group for all $n \geq 2$.

Proposition 34.3.2
Let X be a simply connected space. Let $P \subseteq \mathbb{N}$ be a set of prime numbers. Then the following are equivalent.

- X is P-local.
- $H_n(X, *)$ is P-local for all $n \geq 0$.
- $H_n(\Omega X, *)$ is P-local for all $n \geq 0$.

Lemma 34.3.3
Let A be an abelian group. Let $n \geq 1$. Let $P \subseteq \mathbb{N}$ be a set of prime numbers. If A is P-local, then we have

$$H_*(X; \mathbb{Z}/p\mathbb{Z}) \cong H_*(*, \mathbb{Z}/p\mathbb{Z})$$

for all $p \notin P$.

Definition 34.3.4 (P-Localization)
Let X be a simply connected space. Let $P \subseteq \mathbb{N}$ be a set of prime numbers. A P-localization of X is a space X_P together with a map $i_p : X \to X_P$ such that f induces an isomorphism

$$\pi_n(X) \otimes_{\mathbb{Z}} S_P^{-1}\mathbb{Z} \to \pi_n(X_P)$$

for all $n \geq 2$.

Proposition 34.3.5
Let X be a simply connected space. Let $P \subseteq \mathbb{N}$ be a set of prime numbers. Suppose that X_P exists. Then the following are true.

- Suppose that Z is a P-local space and $f : X \to Z$ is a map. Then there exists a map $g : X_P \to Z$ such that $g \circ i_p = f$.
- Suppose that Z is a P-local space and $f_1, f_2 : X \to Z$ are map. Suppose that $g_1, g_2 : X_P \to Z$ are maps such that $g_1 \circ i_p = f_1$ and $g_2 \circ i_p = f_2$. Suppose that $H : X \times I \to Z$ is a homotopy from f_1 to f_2. Then there exists a homotopy $K : X_P \times I \to Z$ such that $K \circ i_p \times \text{id}_Z = H$.

Proposition 34.3.6
Let X be a simply connected space. Let $P \subseteq \mathbb{N}$ be a set of prime numbers. Then X_P exists and is unique up to homotopy equivalence relative to X.

34.4 Rational Homotopy Equivalence

Definition 34.4.1 (Rational Model)
Let X be a simply connected space. A rational model of X is a P-localization of X where $P = \emptyset$. It is denoted by $X_{\mathbb{Q}}$.

Proposition 34.4.2
Let X be a simply connected space. Then the following are true.

- $X_{\mathbb{Q}}$ is homotopy equivalent to a CW complex.
- $C_*(X_{\mathbb{Q}})$ has all differentials as 0.
- $H_*(X_{\mathbb{Q}})$ is a free graded $\mathbb{Z}$-module.

35 Topics in Simplicial Homology

35.1

Proposition 35.1.1
Let S be a simplicial complex. Let $\sigma \in S$. Let $p \in |\sigma|^\circ$. Then we have

$$H^k_{\text{sing}}(|S|, |S| \setminus p) \cong \widetilde{H}^{k-|\sigma|}_{\Delta}(\text{lk}(\sigma))$$

36 Newtonian Mechanics

We begin with the assumption that we live in a smooth manifold of dimension n, together with an affine connection.

Definition 36.0.1 (Inertial Frame of References)
Let M be a smooth manifold of dimension n. Let ∇ be an affine connection on M. Let (U, φ) be a chart of M. We say that φ is an inertial frame of reference if

$$\frac{d^2(\varphi \circ \gamma)}{dt^2} = 0$$

for all geodesics $\gamma : I \to U$

Lemma 36.0.2
Let M be a smooth manifold of dimension n. Let ∇ be an affine connection on M. If M admits an inertial frame of reference, then ∇ is flat and torsion-free.

Proposition 36.0.3
Let M be a smooth manifold of dimension n. Let ∇ be an torsion-free affine connection on M. Then M admits a global inertial frame of reference if and only if ∇ is flat and M is simply connected.

Newton's first law: inertial frames exists.

36.1 Galilean Space

Definition 36.1.1 (Galilean Space-Time Structure)
Define a Galilean space-time structure to consist of the following data.
- An affine space $(A, \mathbb{R}^4, +)$ called the universe.
- A linear map $\tau : \mathbb{R}^4 \to \mathbb{R}$ called time.
- An inner product $\langle -, - \rangle : \ker(\tau) \times \ker(\tau) \to \mathbb{R}$.

Idea: There are no fixed point of reference in A (no point of origin for both space and time). We measure time relatively by the linear map τ. Elements of $\ker(\tau)$ lies in the same time frame and are called simultaneous events. We measure the spatial difference between simultaneous events by the usual inner product.

Definition 36.1.2 (Galilean Transformations)
Let $(A, \mathbb{R}^4, +, \tau, \langle -, - \rangle)$ and $(A', \mathbb{R}^4, +, \tau', [-, -])$ be Galilean spaces. Let $T : A \to A'$ be an affine transformation. We say that T is a Galilean transformation if the following are true.
- $\tau'(\overline{T}(b) - \overline{T}(a)) = \tau(b - a)$ for all $a, b \in \mathbb{R}^4$.
- $[\overline{T}(b), \overline{T}(a)] = \langle a, b \rangle$ for all $a, b \in \ker(\tau)$.

Definition 36.1.3 (Galilean Isomorphism)
Let $(A, \mathbb{R}^4, +, \tau, \langle -, - \rangle)$ and $(A', \mathbb{R}^4, +, \tau', [-, -])$ be Galilean spaces. Let $T : A \to A'$ be a Galilean transformation. We say that T is an isomorphism if there exists a Galilean transformation S such that

$$S \circ T = \mathrm{id}_A \quad \text{and} \quad T \circ S = \mathrm{id}_{A'}$$

Proposition 36.1.4
Let $(A, \mathbb{R}^4, +, \tau, \langle -, - \rangle)$ be a Galilean space. Then A is isomorphic to $\mathbb{R}^4$ as a Galilean space.

Definition 36.1.5 (The Galilean Group)

Let $(A, \mathbb{R}^4, +, \tau, \langle -, - \rangle)$ be a Galilean space. Define the Galilean group of A to be the subgroup

$$\text{Gal}(A) = \{T : A \to A \mid T \text{ is a Galilean transformation } \} \leq \text{Aff}(A)$$

Definition 36.1.6 (The Galilean Group for the Standard Galilean Space)
Define the Galilean group for $\mathbb{R}^4$ by

$$\text{Gal}(3) = \{T : \mathbb{R}^4 \to \mathbb{R}^4 \mid T \text{ is a Galilean transformation } \} \leq \text{Aff}(\mathbb{R}^4)$$

Let $T \in \text{Gal}(3)$.

- We say that T is uniform motion if T is of the form

  $$T(x, t) = (x + vt, t)$$

  for some $v \in \mathbb{R}^3$.
- We say that T is a translation if T is of the form

  $$T(x, t) = (x + s_0, x + s_1)$$

  for some $s_0 \in \mathbb{R}^3$ and $s_1 \in \mathbb{R}$.
- We say that T is a rotation if T is of the form

  $$T(x, t) = (Gx, t)$$

  for some $G \in O(3, \mathbb{R})$.

Decompose $\text{Gal}(3)$ into rotation + boost + translation

Proposition 36.1.7
Let $T \in \text{Gal}(3)$. There exists a unique rotation $g_1 \in \text{Gal}(3)$, a unique translation $g_2 \in \text{Gal}(3)$ and a unique uniform motion $g_3 \in \text{Gal}(3)$ such that

$$T = g_1 \circ g_2 \circ g_3$$

Proposition 36.1.8
Let $(A, \mathbb{R}^4, +, \tau, \langle -, - \rangle)$ be a Galilean space. Then the following are true regarding inertial frames of references.

- A admits an inertial frame of reference.
- Let ϕ be an inertial frame of reference. Let $T \in \text{Gal}(A)$ be a Galilean transformation. Then $\phi \circ T$ is an inertial frame of reference.
- Let ϕ and ψ be two inertial frames of references. Then there exists a unique $T \in \text{Gal}(A)$ such that $\psi = \phi \circ T$.

36.2 Newton's Laws and Equation of Motion

Definition 36.2.1 (World Lines)
Let $(A, \mathbb{R}^4, +, \tau, \langle -, - \rangle)$ be a Galilean space. Let $r : \mathcal{C}^\infty(\mathbb{R}, \mathbb{R}^3)$ be a map. We say that r is a world line if there exists an inertial frame of reference $\varphi : A \to \mathbb{R}^4$ and a geodesic $\gamma : \mathbb{R} \to A$ such that

$$r = \pi \circ \varphi \circ \gamma$$

where $\pi : \mathbb{R}^4 \to \mathbb{R}^3$ is the projection map to the first three coordinates.

Definition 36.2.2 (Newtonian System)
Let $(A, \mathbb{R}^4, +, \tau, \langle -, - \rangle)$ be a Galilean space. Let $N \in \mathbb{N}^+$. A Newtonian system of N objects consists of the following data.

- World lines $r_1, \dots, r_N \in \mathcal{C}^\infty(\mathbb{R}, \mathbb{R}^3)$.

- Masses $m_1, \ldots, m_N \in \mathbb{R}^+$.
- Force Laws $F_i : \mathbb{R}^{3N} \times \mathbb{R}^{3N} \times \mathbb{R} \to \mathbb{R}$ for $1 \leq i \leq N$.

such that
$$m_i \frac{d^2 r_i}{dt^2} = F_i \left(r_1, \ldots, r_N, \frac{dr_1}{dt}, \ldots, \frac{dr_N}{dt}, t \right)$$
for $1 \leq i \leq N$.

The condition that F_i is constant ration with r'' is Newton's second law.

Postulate: Newton's principle of determinacy: global solution of of the ode exists given initial data (only local solution is guaranteed by Picard-Lindelof)

Definition 36.2.3 (Galilean Newtonian System)
Let $(A, \mathbb{R}^4, +, \tau, \langle -, - \rangle)$ be a Galilean space. Let $N \in \mathbb{N}^+$. Let $(r_1, \ldots, r_N, m_1, \ldots, m_N, F_1, \ldots, F_N)$ be a Newtonian system of N objects. We say that the Newtonian system of N objects is Galilean if
$$F_i \left(Gr_1 + vt + s_0, \ldots, Gr_N + vt + s_0, G\frac{dr_1}{dt} + v, \ldots, G\frac{dr_N}{dt} + v, t + s_1 \right) = F_i \left(r_1, \ldots, r_N, \frac{dr_1}{dt}, \ldots, \frac{dr_N}{dt}, t \right)$$
for all $1 \leq i \leq N$ and all $G \in O(3)$, $v, s_0 \in \mathbb{R}^3$ and $s_1 \in \mathbb{R}$.

Postulate: Galileo's principle of relativity. All observed forces are Galilean. (All Newtonian systems are Galilean)

36.3 Gravitational Force

Definition 36.3.1 (System for Gravitational Force)
The system for gravitational force is a Newtonian system of 2 objects whose force law is given by
$$F_1 = \frac{Gm_1m_2}{|r_2 - r_1|^3}(r_2 - r_1) \quad \text{and} \quad F_2 = \frac{Gm_1m_2}{|r_1 - r_2|^3}(r_1 - r_2)$$
where G is the gravitational constant.

Example 36.3.2
If the first object is a stone and the second object is earth, then $m_2 >> m_1$. Then setting $F_1 = m_1 \frac{d^2 r_1}{dt^2}$ and $F_2 = m_2 \frac{d^2 r_2}{dt^2}$, we deduce that
$$\frac{d^2 r_2/dt^2}{d^2 r_1/dt^2} = \frac{m_1}{m_2} \sim 0$$
This implies that and so $r_2(t) = k$ is approximately constant. Then the equation for F_1 reduces to
$$\frac{d^2 r_1}{dt^2} = -\frac{Gm_2}{|r_1|^3} r_1$$
Assuming that the stone is close to the earth's surface, we have $|r_1| \sim R$ where R is the earth's radius. Write $r_1(t) = (x(t), y(t), z(t))$. In an appropriate inertial frame, we may decouple the equations to get
$$\frac{d^2 x}{dt^2} = 0 \quad \text{and} \quad \frac{d^2 y}{dt^2} = 0 \quad \text{and} \quad \frac{d^2 z}{dt^2} = -g$$
where $g = \frac{Gm_2}{R^2}$. Assuming that the stone is dropped from height h_0. Then solving the second order ODEs with the initial data gives
$$r_1(t) = \left(0, 0, h_0 - \frac{1}{2} gt \right)$$
The stone hits the ground when $t = \sqrt{2h_0/g}$.

37 Lagrangian Mechanics

37.1 Configuration Space

Definition 37.1.1 (Holonomic Constraints)
Let $N \in \mathbb{N}^+$. Let $(r_1, \dots, r_N, m_1, \dots, m_N, F_1, \dots, F_N)$ be a Newtonian system of N objects. A holonomic constraint on the system is a function $f \in \mathcal{C}^2(\mathbb{R}^{3N} \times \mathbb{R}, \mathbb{R})$ such that

$$f(r_1, \dots, r_N, t) = 0$$

Definition 37.1.2 (Configuration Space)
Let $N \in \mathbb{N}^+$. Let $(r_1, \dots, r_N, m_1, \dots, m_N, F_1, \dots, F_N)$ be a Newtonian system of N objects. Let $f_1, \dots, f_k$ be holonomic constrains on the system. Define the configuration space of the system to be

$$Q = \{x \in \mathbb{R}^{3N} \mid f_j(x) = 0 \text{ for all } 1 \leq j \leq N\}$$

Each point in the configuration space is not a point in spacetime, instead they represent a possible position for N particles to situate in space time.

Definition 37.1.3 (Trajectories)
Let Q be a configuration space of N objects. A trajectory on Q is a smooth function

$$\gamma : [t_1, t_2] \subseteq \mathbb{R} \to Q$$

Definition 37.1.4 (The Space of Trajectories)
Let Q be a configuration space of N objects. Define the space of trajectories of Q to be

$$\Omega Q = \{\gamma \mid \gamma \text{ is a trajectory }\}$$

37.2 The Euler Lagrange Operator

Definition 37.2.1 (Lagrangian)
Let Q be a configuration space of N objects. A Lagrangian on Q is a smooth function

$$L : TQ \times \mathbb{R} \to \mathbb{R}$$

Definition 37.2.2 (The Action Functional)
Let Q be a configuration space of N objects. Let $L : TQ \times \mathbb{R} \to \mathbb{R}$ be a Lagrangian. Let $\gamma : [t_1, t_2] \to Q$ be a trajectory. Define the action functional of γ by

$$S[\gamma] = \int_{t_1}^{t_2} L(\gamma(t), \gamma'(t), t) \, dt$$

Definition 37.2.3 (Space of Deformations)
Let Q be a configuration space of N objects. Let $\gamma : [t_1, t_2] \to Q$ be a trajectory. Define the space of deformation of γ by

$$\Omega_\gamma = \{\eta : [t_1, t_2] \to TQ \mid \eta(t) \in T_{\gamma(t)}Q \text{ and } \eta(t_1) = \eta(t_2) = 0\}$$

Definition 37.2.4 (Variations)
Let Q be a configuration space of N objects. Let $L : TQ \times \mathbb{R} \to \mathbb{R}$ be a Lagrangian. Let $\gamma : [t_1, t_2] \to Q$ be a trajectory. Let $\eta \in \Omega_\gamma$ be a deformation. Define the variation of η to be

$$\delta S[\gamma](\eta) = \frac{d}{d\varepsilon}\bigg|_{\varepsilon=0} S[\gamma + \varepsilon\eta]$$

Assumption: Hamilton's principle (principle of least / stationary action): the actual path taken by the system is the one for which $S[\gamma]$ is stationary under small perturbations (i.e. $\delta S[\gamma](\eta) = 0$ for all η)

Definition 37.2.5 (Euler-Lagrange Operator)
Let Q be a configuration space of N objects. Let $L : TQ \times \mathbb{R} \to \mathbb{R}$ be a Lagrangian. Define the Euler Lagrange operator to be the function $\text{EL}^i : \Omega \to \mathcal{C}^0([t_1, t_2], \mathbb{R}^n)$ by

$$\text{EL}^i(\gamma)(t) = \frac{\partial L}{\partial x^i}\bigg|_{\gamma(t)} - \frac{d}{dt}\frac{\partial L}{\partial x_i'}\bigg|_{\gamma(t)}$$

for any $\gamma \in \Omega$ where $(x_1', \dots, x_n', x^1, \dots, x^n)$ are the coordinates of TQ.

Example 37.2.6
Suppose that $Q = \mathbb{R}^1$ is the configuration space of 1 object. Let $L : TQ \times \mathbb{R} \to \mathbb{R}$ be the Lagrangian given by

$$L(v, x, t) = \frac{1}{2}mv^2 - V(x)$$

Then for any $\gamma : [0, 1] \to Q$, the Euler Lagrange equation operator reads

$$\text{EL}(\gamma) = -V'(\gamma(t)) - m\gamma''(t)$$

Notice that setting $\text{EL}(\gamma)$ recovers Newton's second equation of motion where $F = -V'(\gamma(t))$.

Theorem 37.2.7
Let Q be a configuration space of N objects. Let $L : TQ \times \mathbb{R} \to \mathbb{R}$ be a Lagrangian. Let $\gamma : [t_1, t_2] \to Q$ be a trajectory. Then we have $\delta S[\gamma](\eta) = 0$ for all $\eta \in \Omega_\gamma$ if and only if $\text{EL}^i(\gamma) = 0$ for all i.

37.3 Symmetries and Noether's Theorem

Definition 37.3.1 (Infinitesimal Symmetry)
Let Q be a configuration space of N objects. Let $L : TQ \times \mathbb{R} \to \mathbb{R}$ be a Lagrangian. Let $\Phi : (-\varepsilon, \varepsilon) \times Q \to Q$ be a one parameter family of diffeomorphisms. We say that Φ is a symmetry of L if

$$L(\Phi_s, (T\Phi_s)(q'), t) = L(q, q', t)$$

for all $s \in (-\varepsilon, \varepsilon)$.

Definition 37.3.2 (Infinitesimal Generator)
Let Q be a configuration space of N objects. Let $L : TQ \times \mathbb{R} \to \mathbb{R}$ be a Lagrangian. Let $\Phi : (-\varepsilon, \varepsilon) \times Q \to Q$ be a symmetry of L. Define the infinitesimal generator of Φ to be the vector field

$$\xi = \frac{d}{ds}_{s=0}\Phi_s$$

Theorem 37.3.3 (Noether's Theorem)
Let Q be a configuration space of N objects. Let $L : TQ \times \mathbb{R} \to \mathbb{R}$ be a Lagrangian. Let $\Phi : (-\varepsilon, \varepsilon) \times Q \to Q$ be a symmetry of L. Let γ be a solution of EL. Then we have

$$\frac{d}{dt}\frac{\partial L}{\partial x_i'}\xi^i\bigg|_\gamma = 0$$

38 Hamiltonian Mechanics

39 K Theory

39.1

Definition 39.1.1 (The 0th K-Group)
Let R be a ring. Define the 0th K-group of R to be the set

$$K_0(R) = \frac{\{P \mid P \text{ is a finitely generated projective } R\text{-module }\}}{\cong}$$

together with addition given by $[P_1] + [P_2] = [P_3]$ if there is a short exact sequence

$$0 \longrightarrow P_1 \longrightarrow P_3 \longrightarrow P_2 \longrightarrow 0$$

Example 39.1.2
Suppose that R is a PID. Then there is an isomorphism

$$K_0(R) \cong \mathbb{Z}$$

given by $[n] \mapsto [R^n]$.

Definition 39.1.3 (The Change of Rings Homomorphism)
Let R, S be rings. Let $f : R \to S$ be a ring homomrphism. Define the change of rings homomorphism to be the map $K_0(f) : K_0(R) \to K_0(S)$ given by

$$[P] \mapsto [S \otimes_R P]$$

Recall the Grothendieck completion of an abelian semi-group $G(-)$.

Proposition 39.1.4
Let R be a ring. Let $P(R) = \{P \mid P \text{ is a finitely generated projective } R\text{-modules }\}$. Then there is a natural isomorphism

$$K_0(R) \cong G(P(R), \oplus)$$

Proposition 39.1.5 (Morita Equivalence)
Let R be a ring. Then there is a natural isomorphism

$$K_0(R) \cong K_0(M_n(R))$$

given by $[P] \mapsto R^n \otimes_R P$ where R^n is considered as an $(M_n(R), R)$-bimodule.

Proposition 39.1.6
Let R, S be rings. Then there is a natural isomrophism

$$K_0(R \times S) \cong K_0(R) \times K_0(S)$$

induced by the product of the projection maps.

Example 39.1.7
Let F be a field. Let V be a vector space over F of countable dimension. Then there is an isomorphism

$$K_0(\text{End}_F(V)) \cong \{1\}$$

Definition 39.1.8 (The Reduced 0th K-Group)

Let R be a ring. Define the reduced 0th K-group to be the quotient

$$\widetilde{K}_0(R) = \frac{K_0(R)}{\langle [R^m] - [R^n] \mid m, n \in \mathbb{N} \rangle}$$

Lemma 39.1.9
Let R be a ring. Then there is a natural isomorphism

$$\widetilde{K}_0(R) \cong \langle [R] \rangle \leq \widetilde{K}_0(R)$$

Proposition 39.1.10
Let G be a group. Let R be an integral domain. Then there is an isomorphism

$$K_0(R[G]) \cong \widetilde{K}_0(R[G]) \oplus \mathbb{Z}$$

give by $[P] \mapsto ([P], \dim_{\text{Frac}(R)}(\text{Frac}(R) \otimes_{R[G]} P))$

Conjecture 39.1.11 (Farrell-Jones Conjecture)
Let R be a regular ring. Let G be a torsion free group. Then the inclusion map induces a natural isomorphism

$$K_0(R) \cong K_0(R[G])$$